

# BFS Traversal

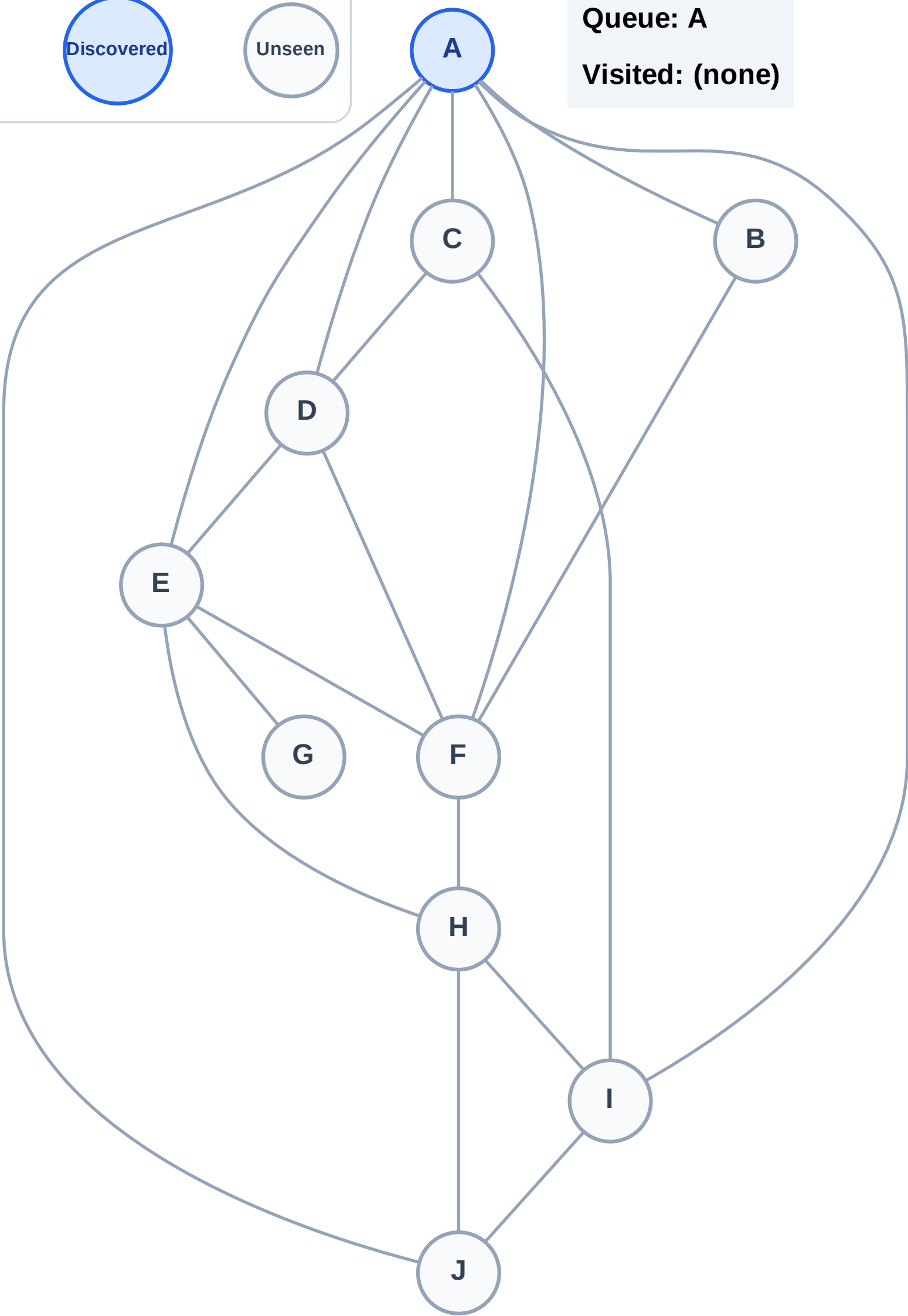
Step 1: Start at A

## Legend



Queue: A

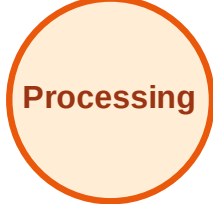
Visited: (none)



# BFS Traversal

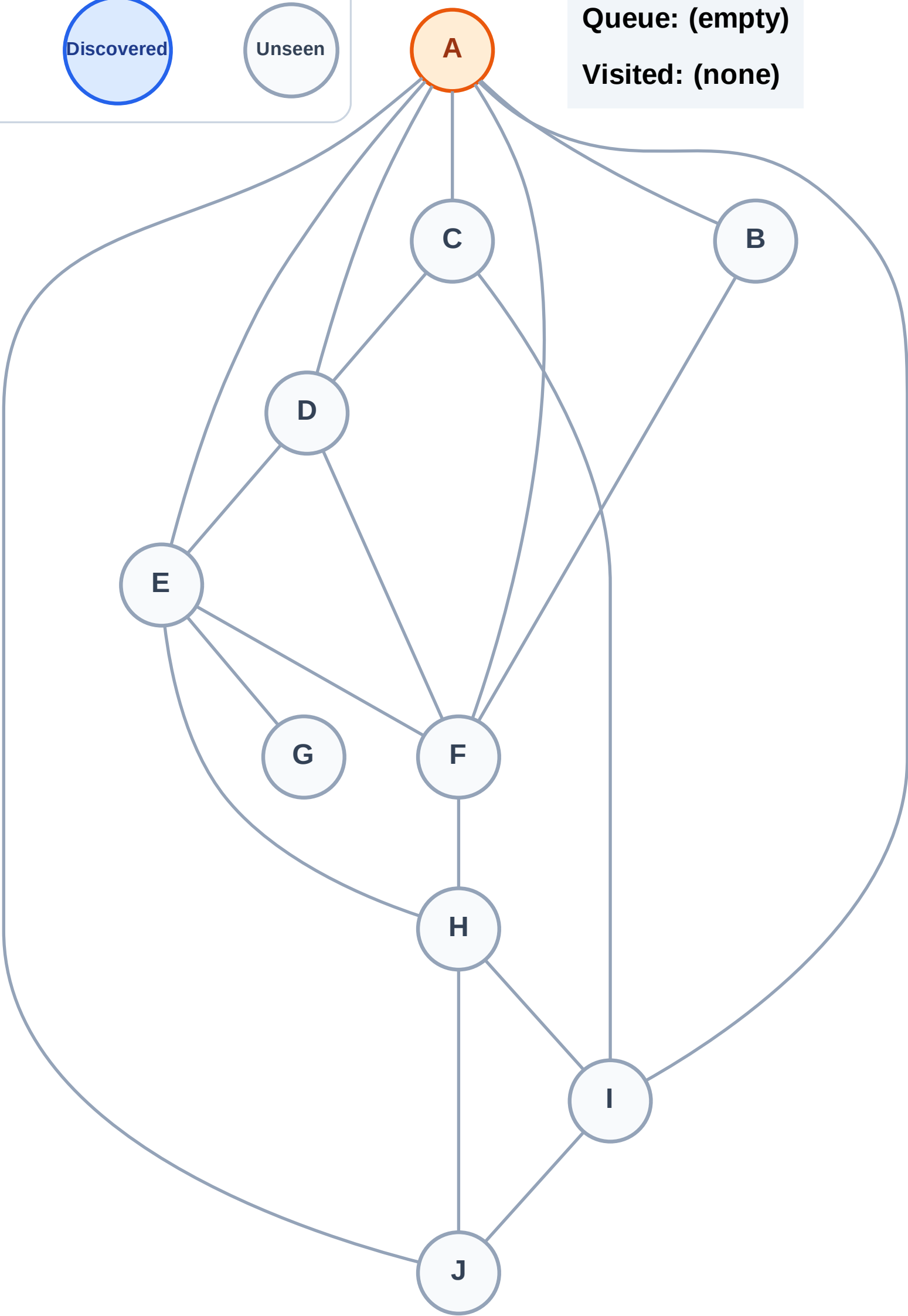
Step 2: Process node A

## Legend



Queue: (empty)

Visited: (none)



# BFS Traversal

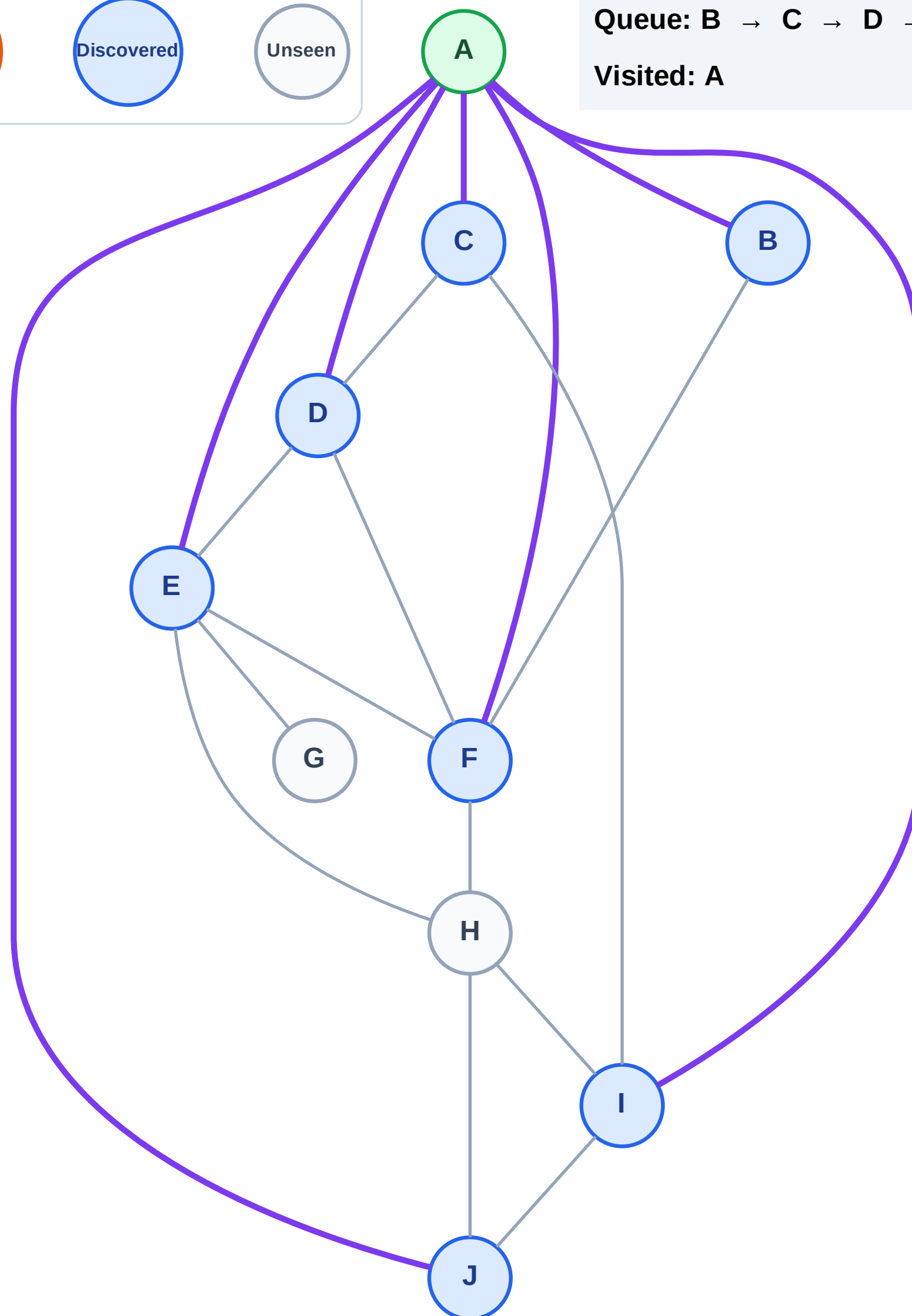
Step 3: Discover B, C, D, E, F, I, J from A

## Legend



Queue: B → C → D → E → F → I → J

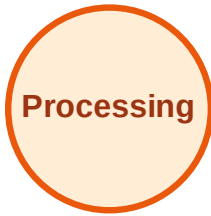
Visited: A



# BFS Traversal

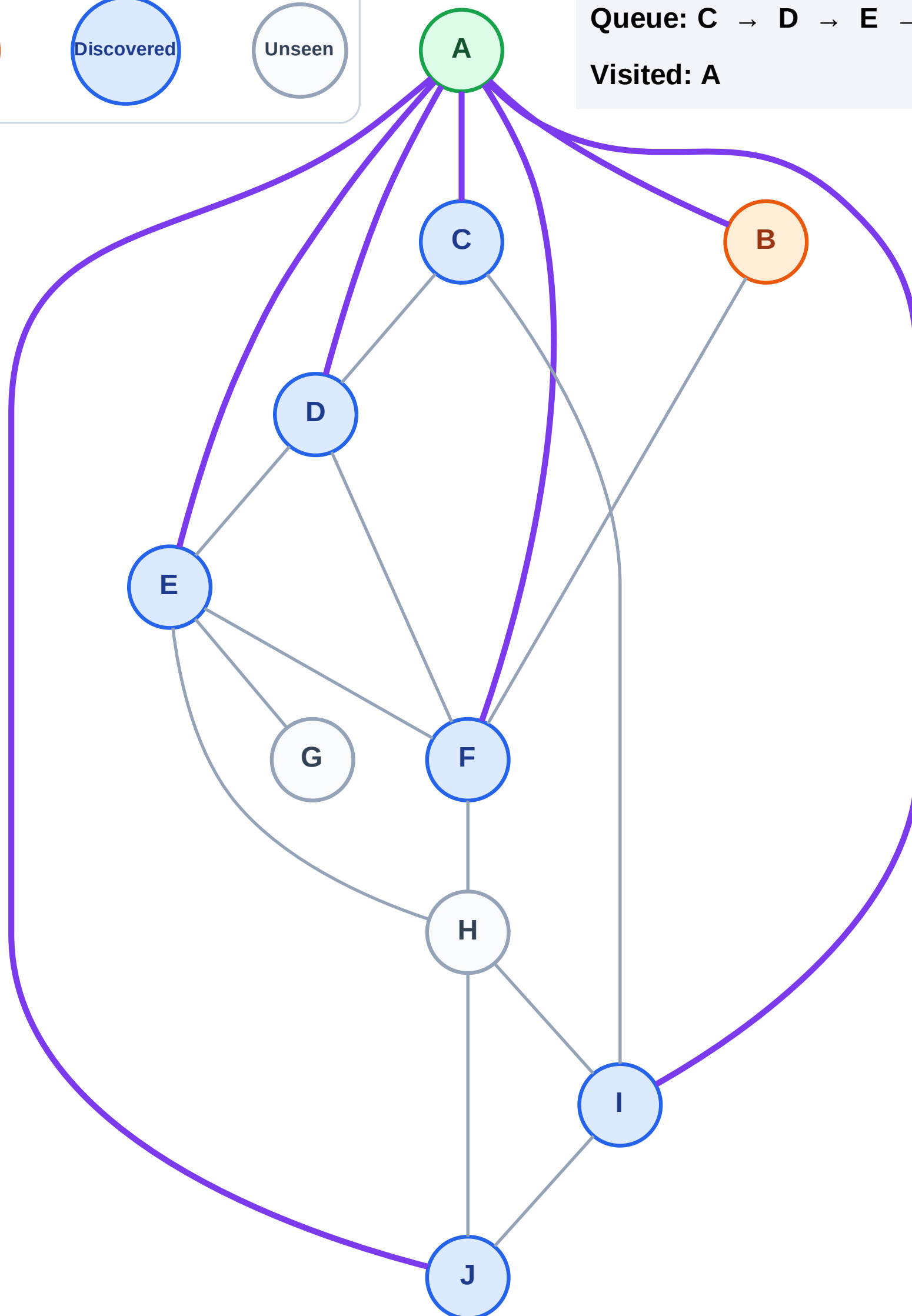
Step 4: Process node B

## Legend



Queue: C → D → E → F → I → J

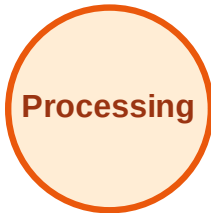
Visited: A



# BFS Traversal

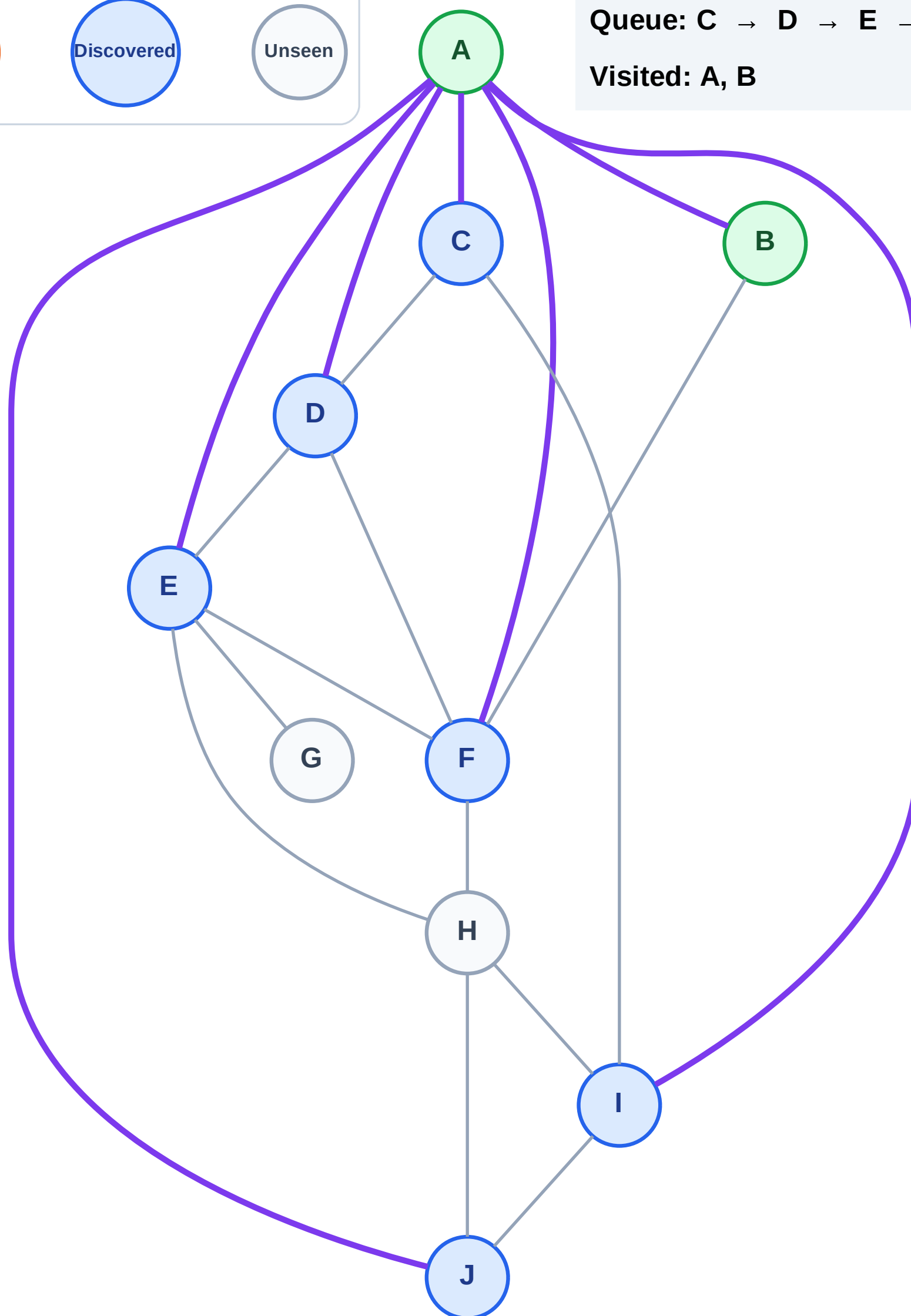
Step 5: No new neighbors from B

## Legend



Queue: C → D → E → F → I → J

Visited: A, B





# BFS Traversal

Step 6: Process node C

Legend

Complete

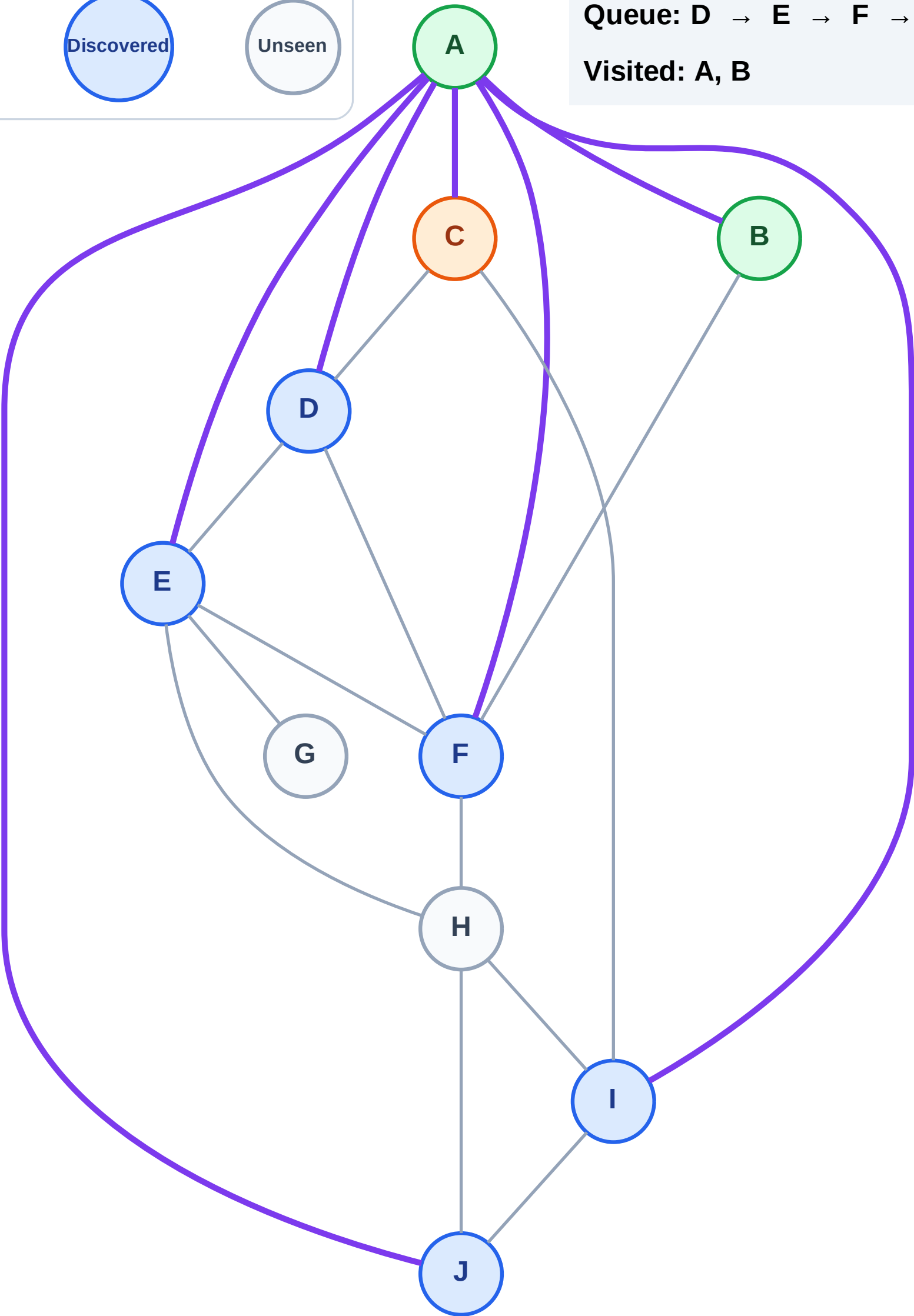
Processing

Discovered

Unseen

Queue: D → E → F → I → J

Visited: A, B



# BFS Traversal

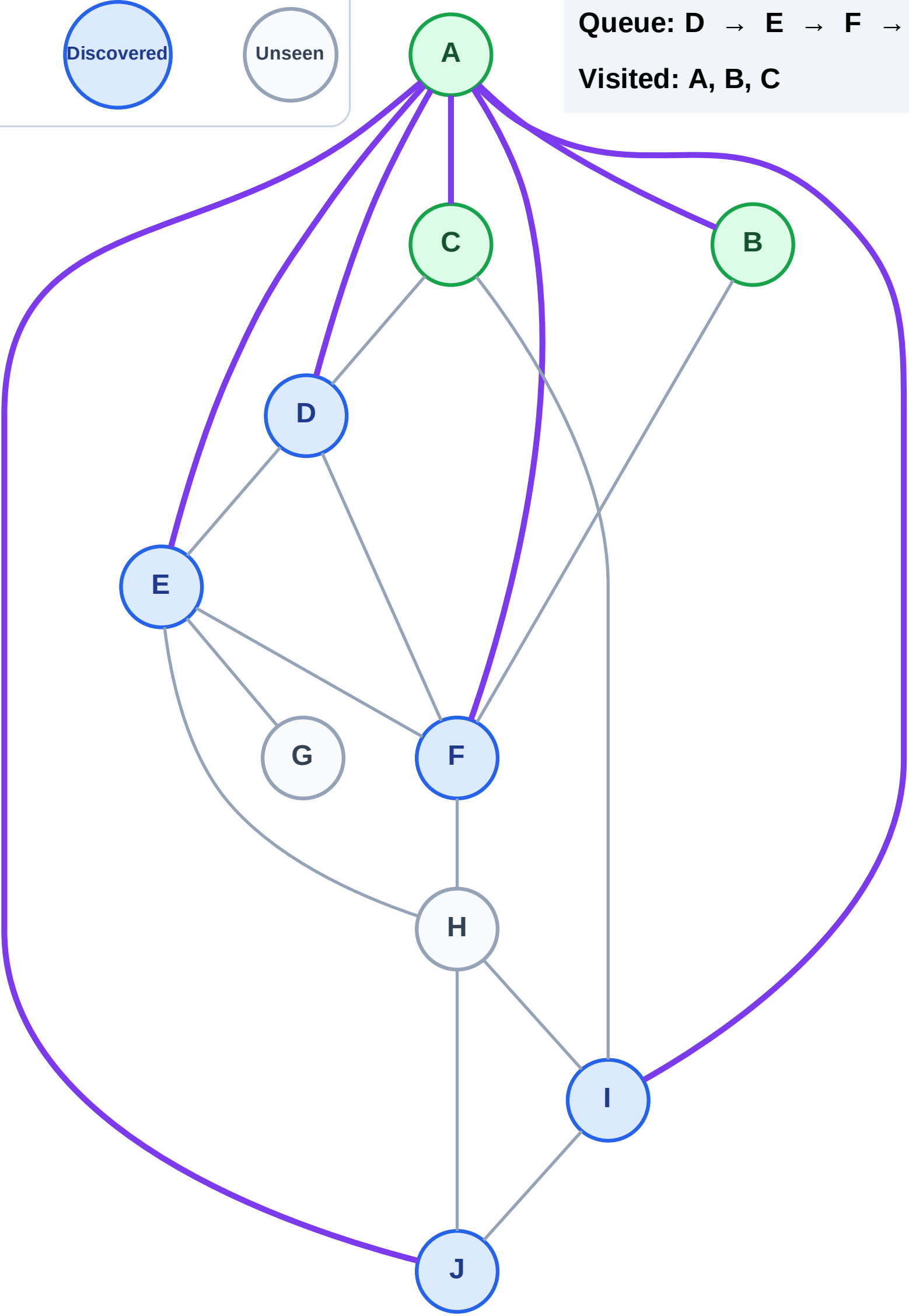
Step 7: No new neighbors from C

## Legend



Queue: D → E → F → I → J

Visited: A, B, C



# BFS Traversal

Step 8: Process node D

Legend

Complete

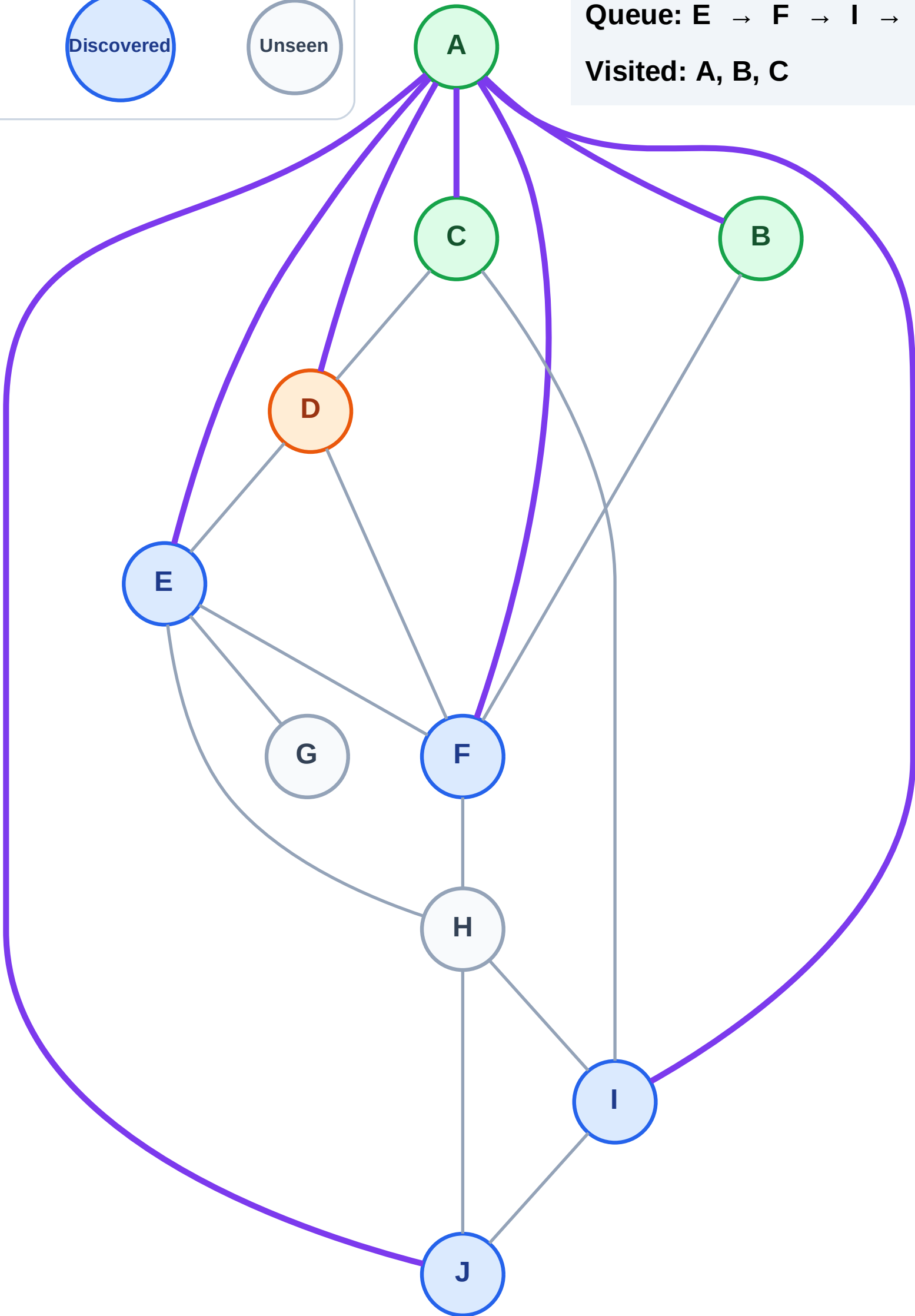
Processing

Discovered

Unseen

Queue: E → F → I → J

Visited: A, B, C





# BFS Traversal

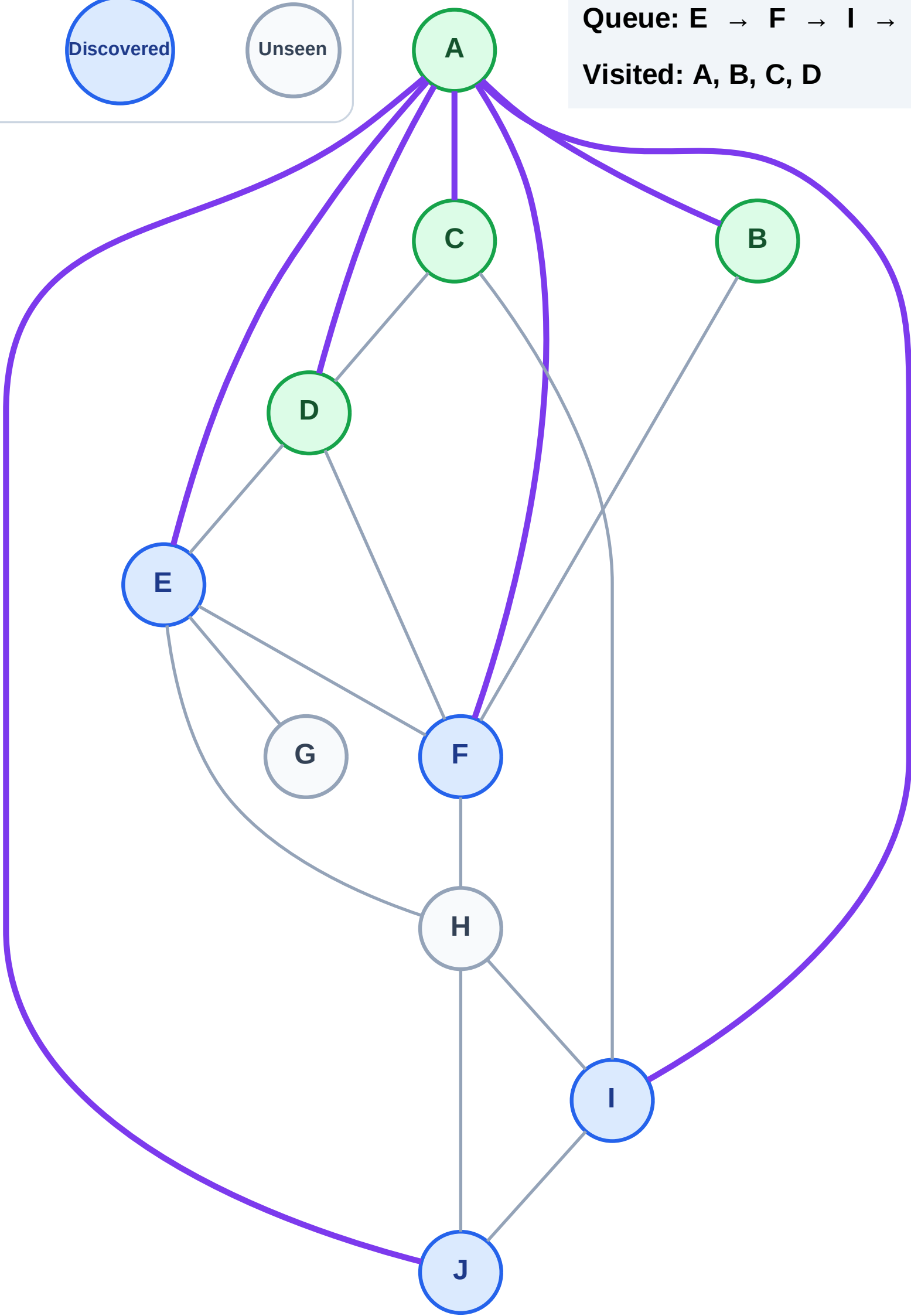
Step 9: No new neighbors from D

## Legend



Queue: E → F → I → J

Visited: A, B, C, D



# BFS Traversal

Step 10: Process node E

Legend

Complete

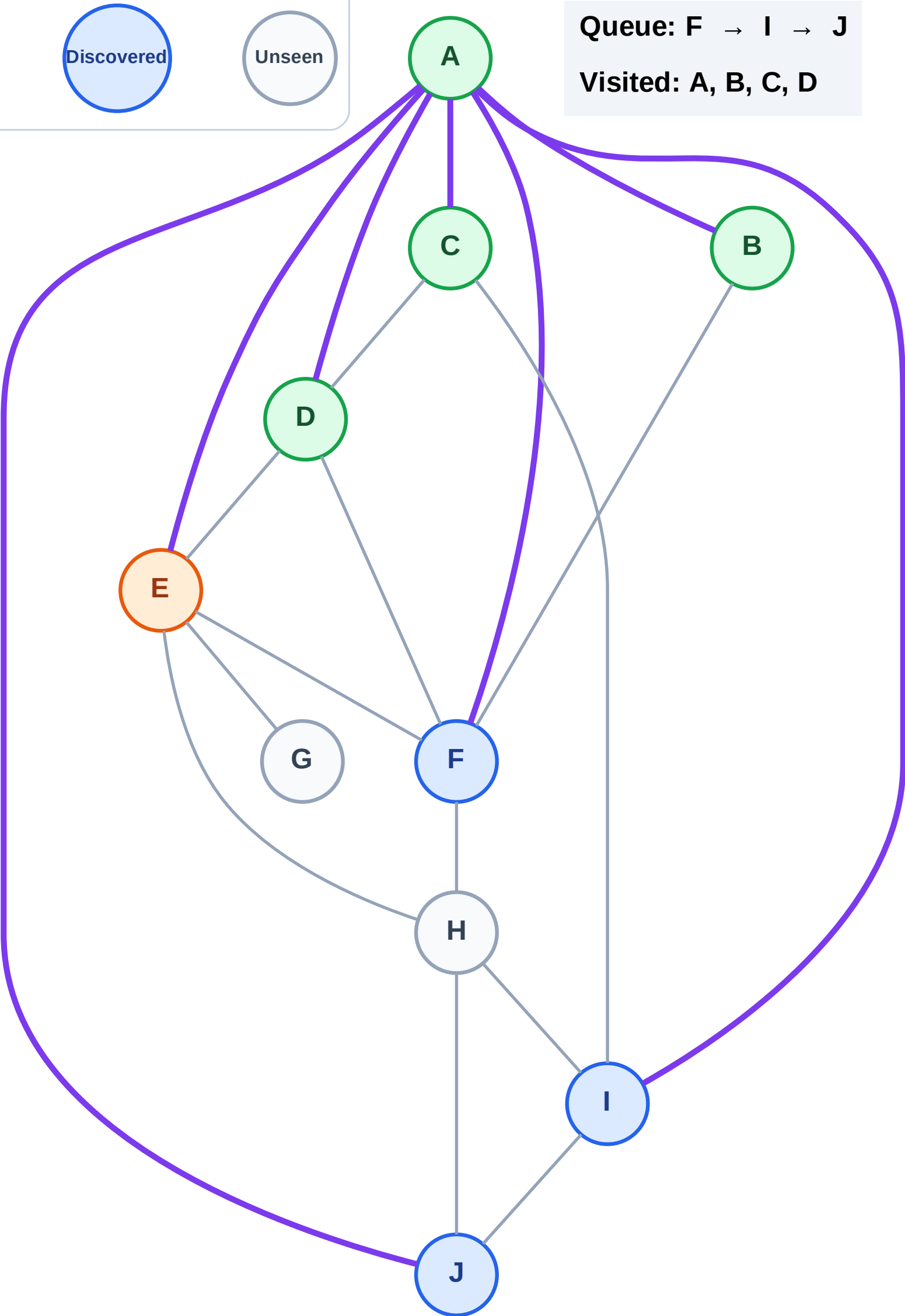
Processing

Discovered

Unseen

Queue: F → I → J

Visited: A, B, C, D



# BFS Traversal

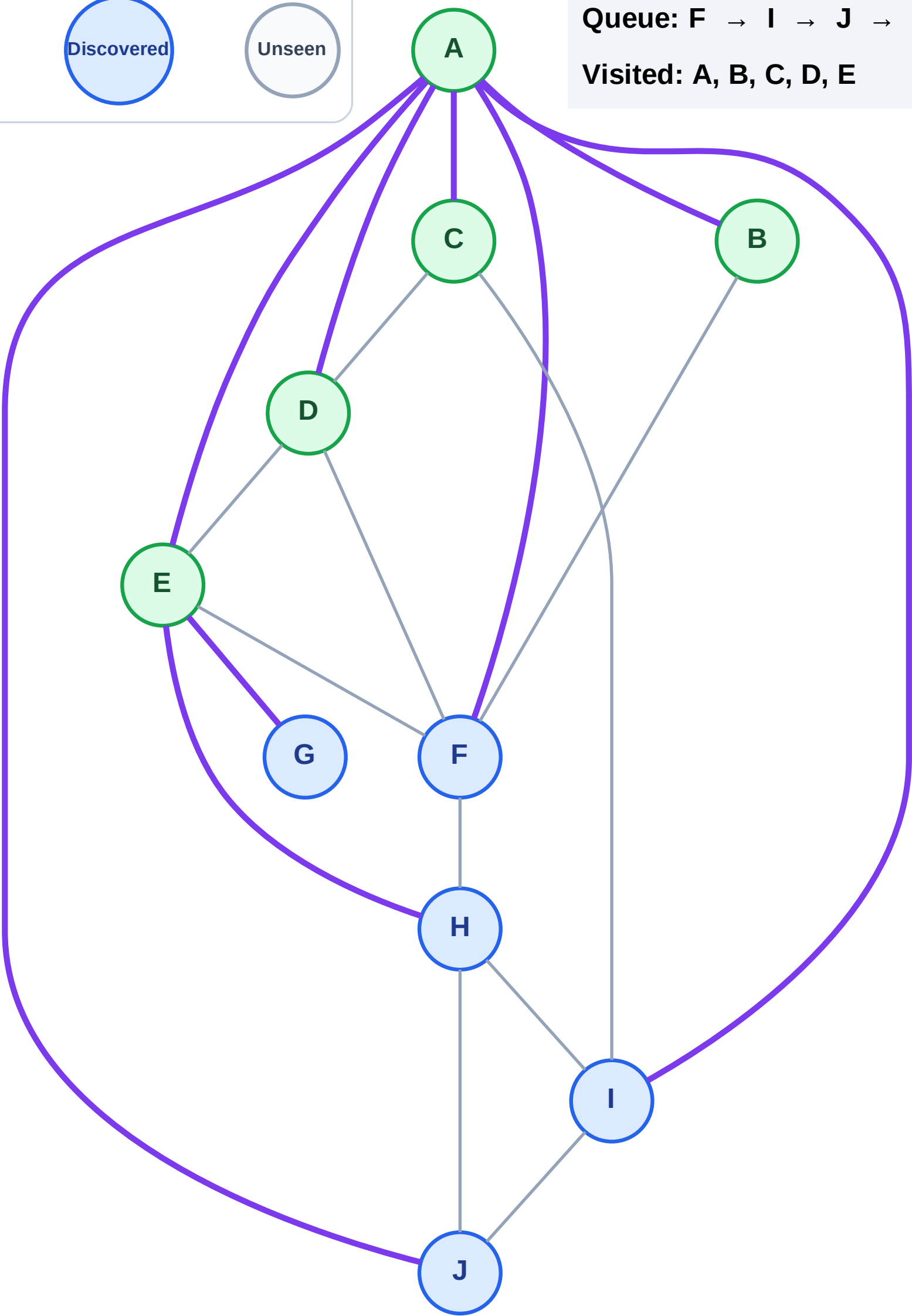
Step 11: Discover G, H from E

## Legend



Queue: F → I → J → G → H

Visited: A, B, C, D, E



# BFS Traversal

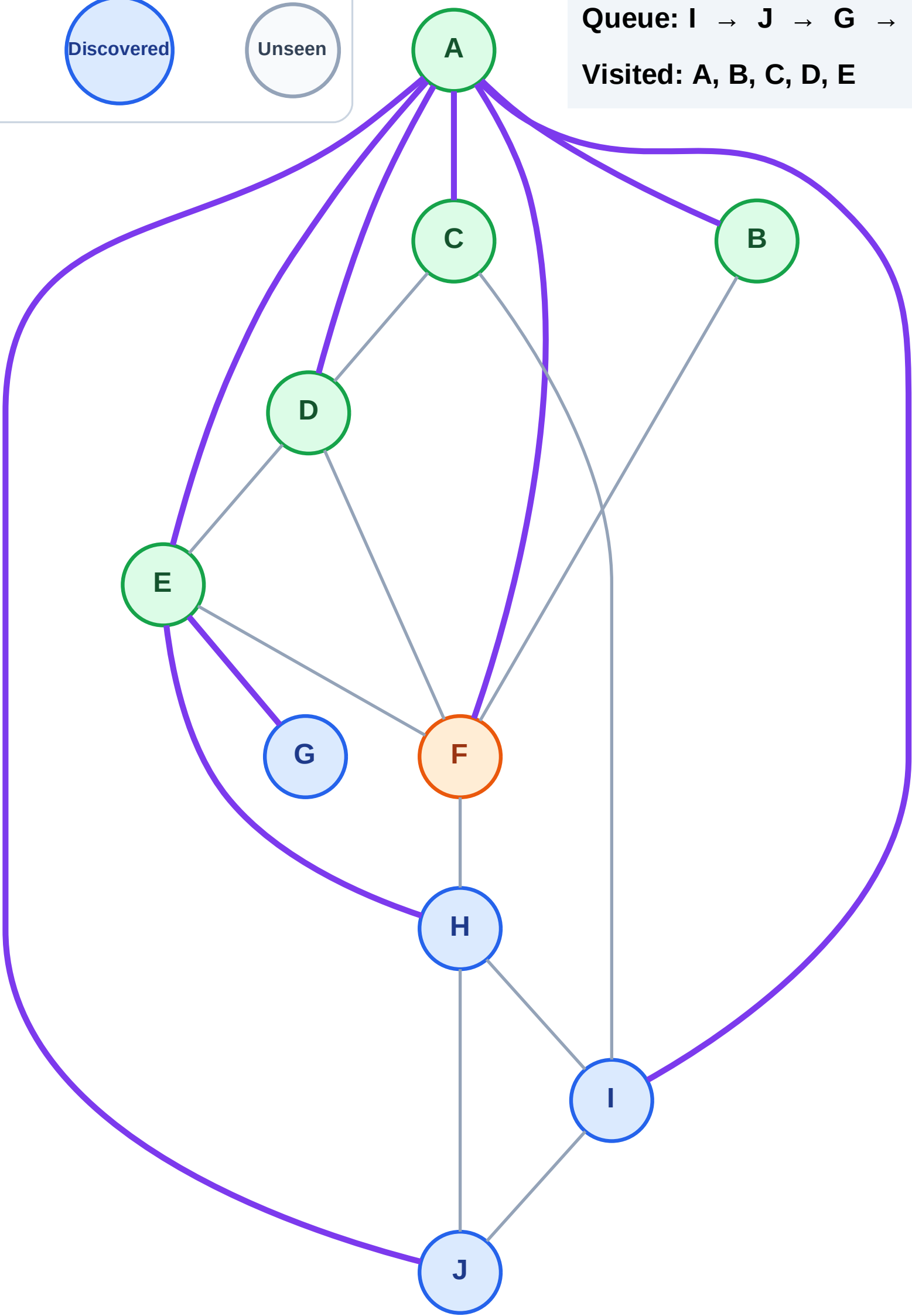
Step 12: Process node F

## Legend



Queue: I → J → G → H

Visited: A, B, C, D, E



# BFS Traversal

Step 13: No new neighbors from F

Legend

Complete

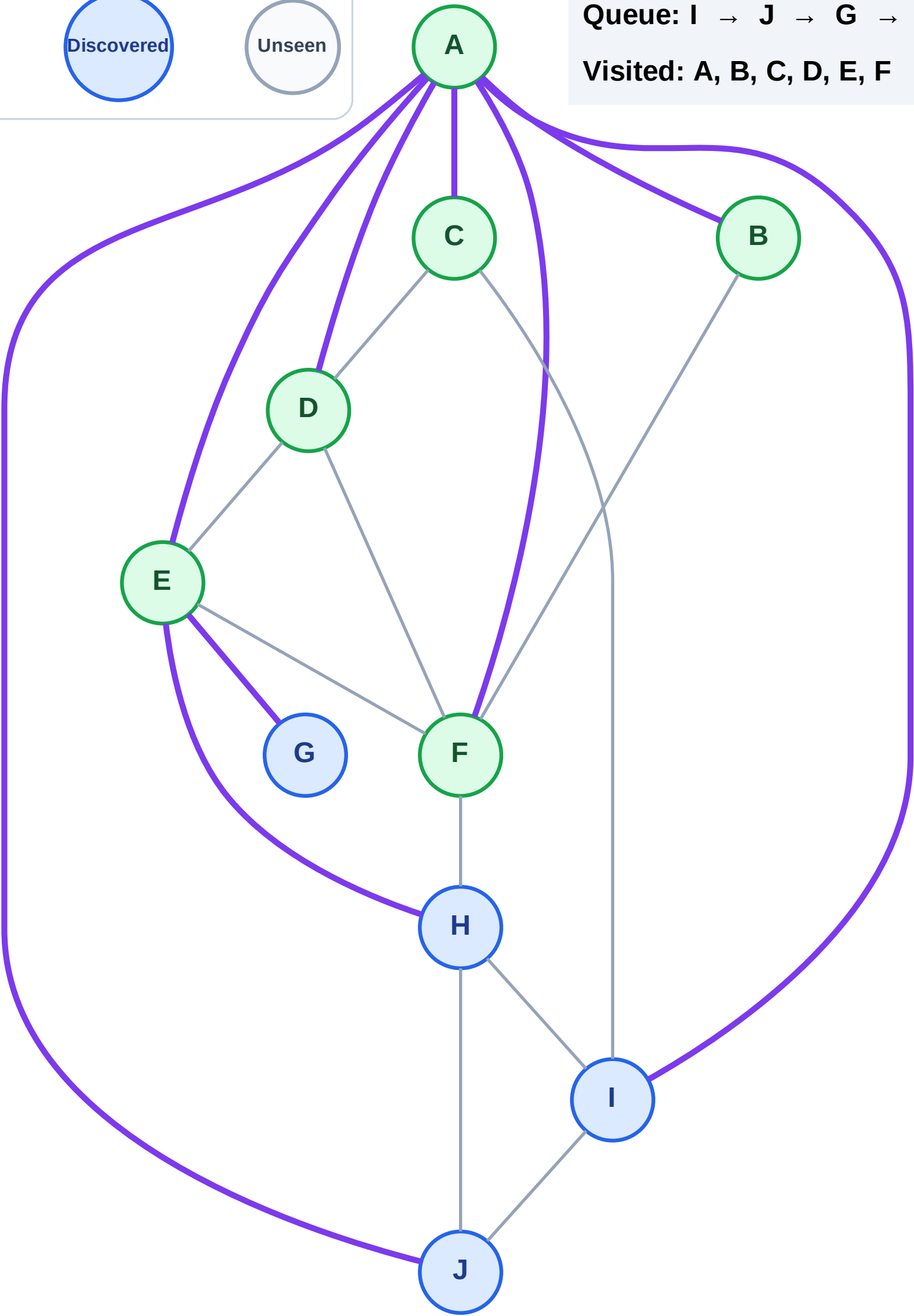
Processing

Discovered

Unseen

Queue: I → J → G → H

Visited: A, B, C, D, E, F



# BFS Traversal

Step 14: Process node I

Legend

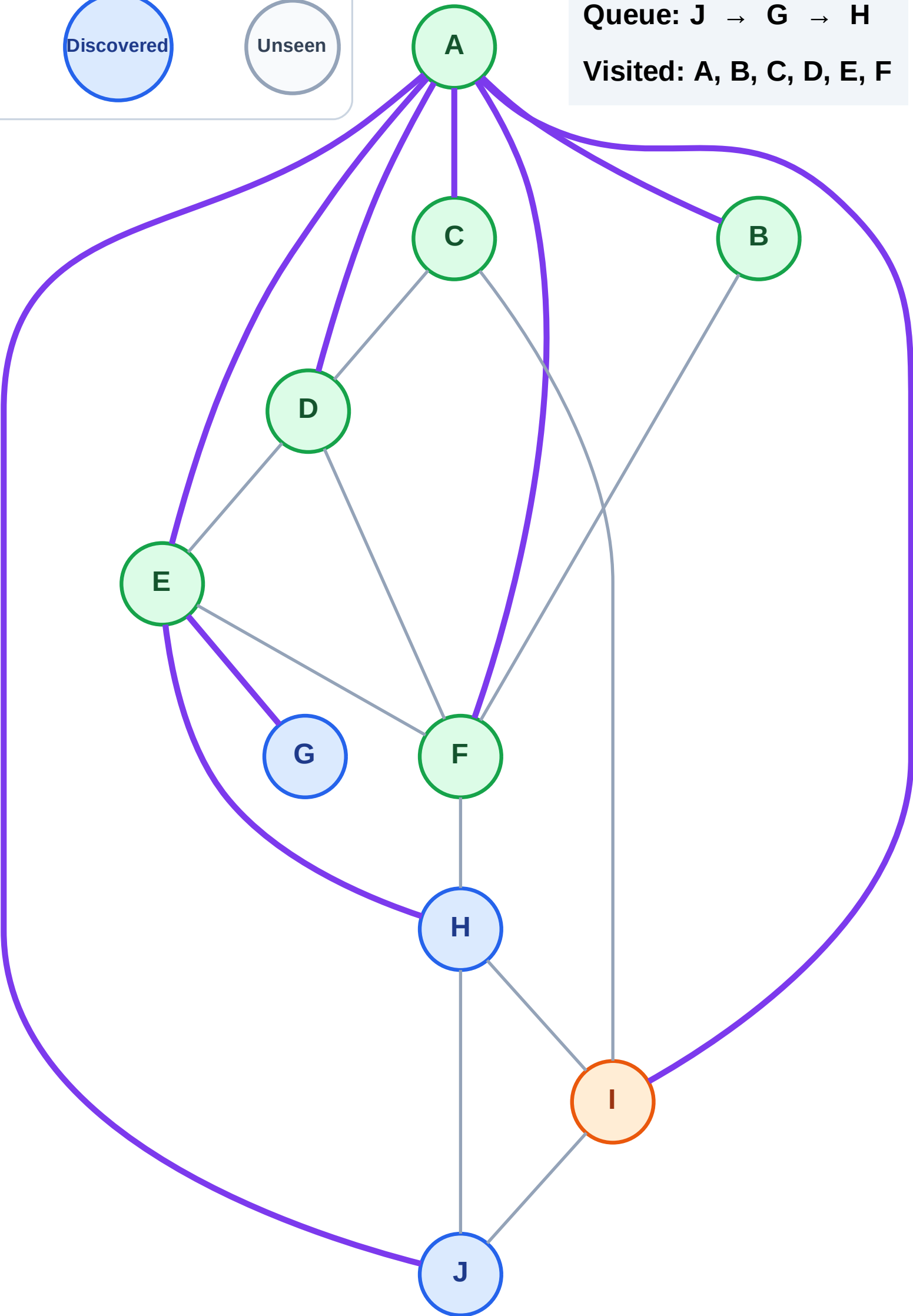
Complete

Processing

Discovered

Unseen

Queue: J → G → H  
Visited: A, B, C, D, E, F





# BFS Traversal

Step 15: No new neighbors from I

## Legend

Complete

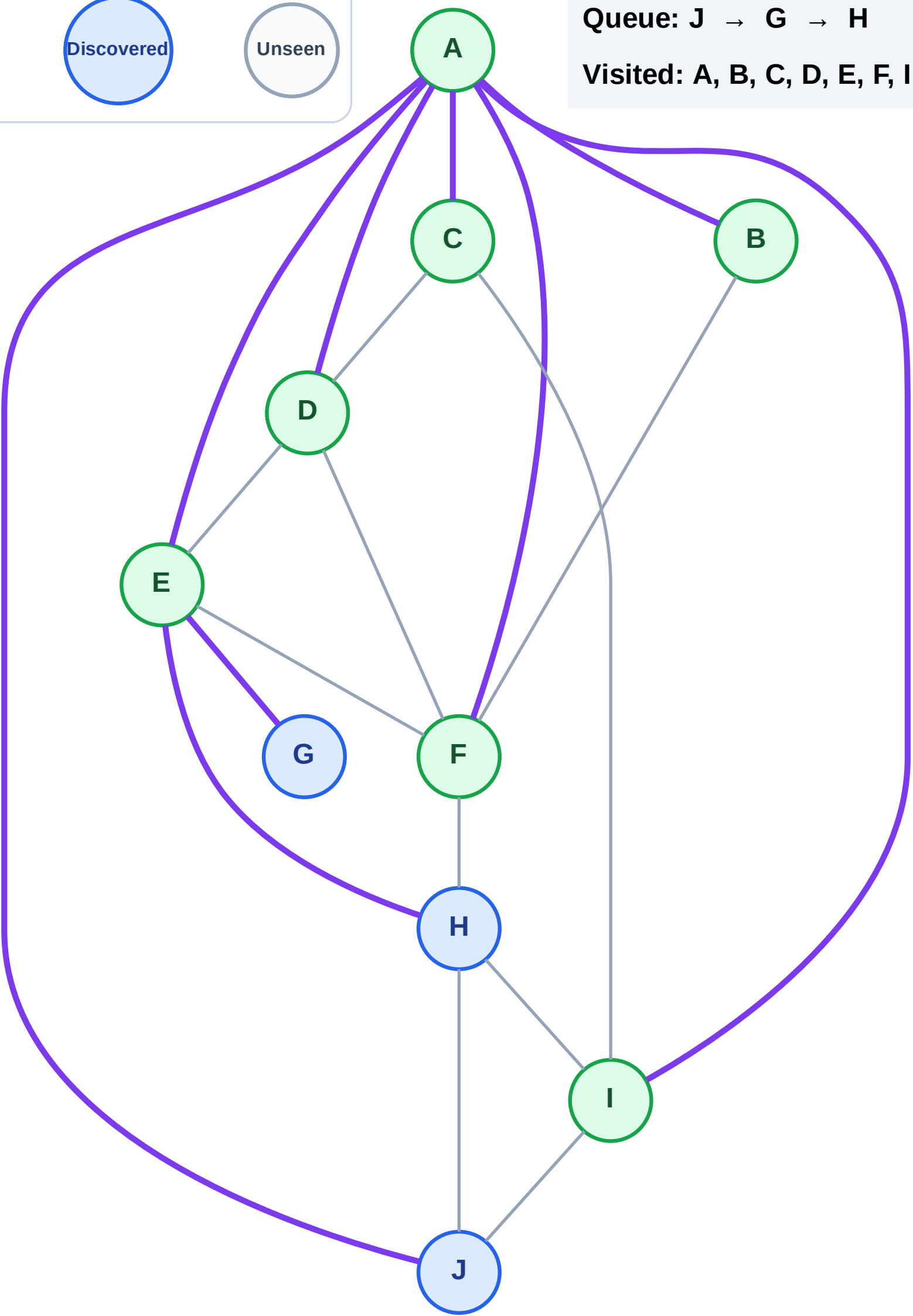
Processing

Discovered

Unseen

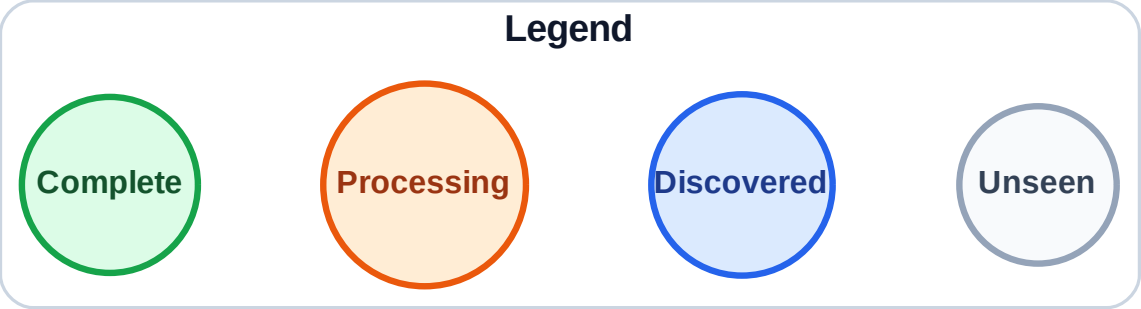
Queue: J → G → H

Visited: A, B, C, D, E, F, I

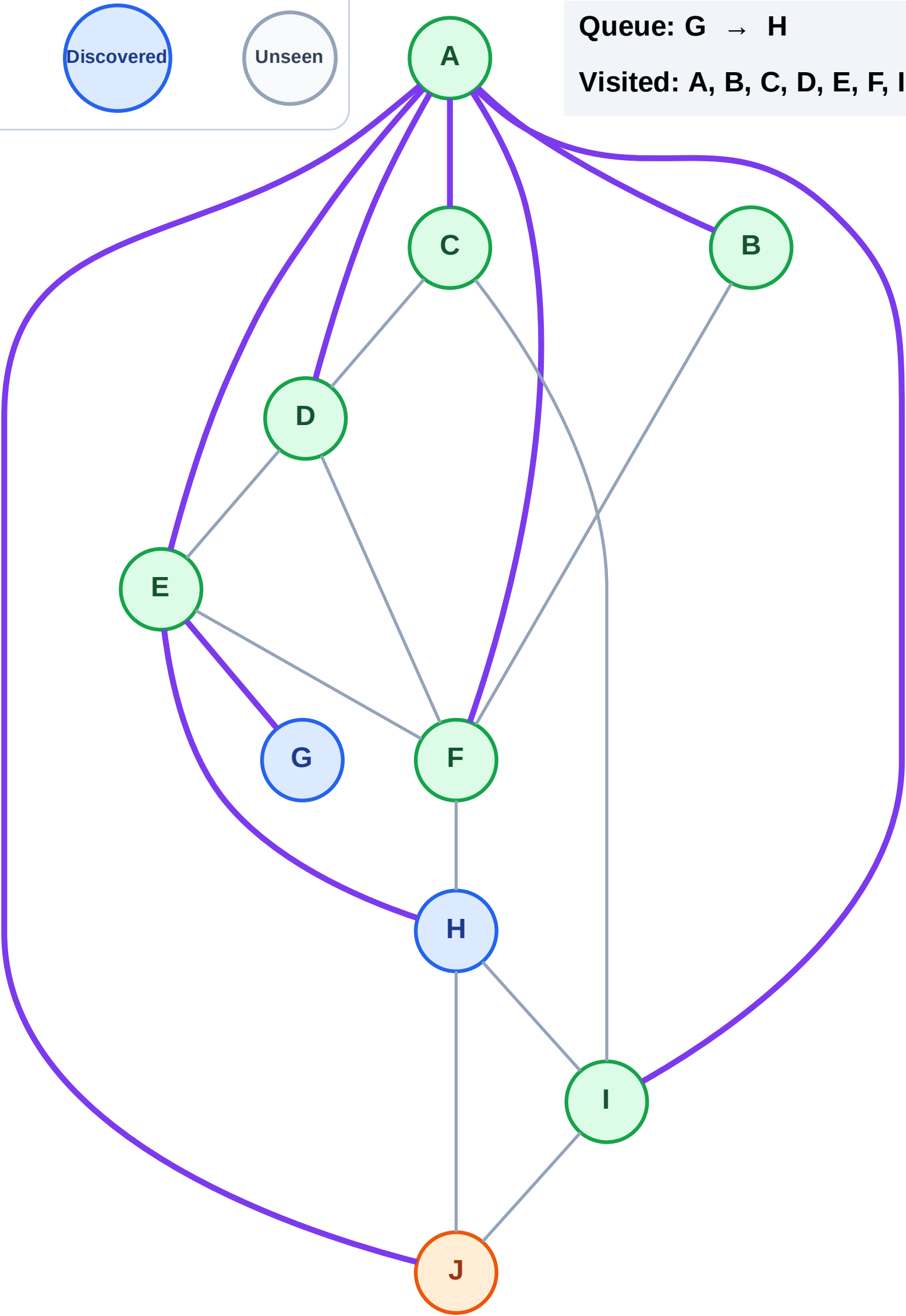


# BFS Traversal

Step 16: Process node J



Queue: G → H  
Visited: A, B, C, D, E, F, I





# BFS Traversal

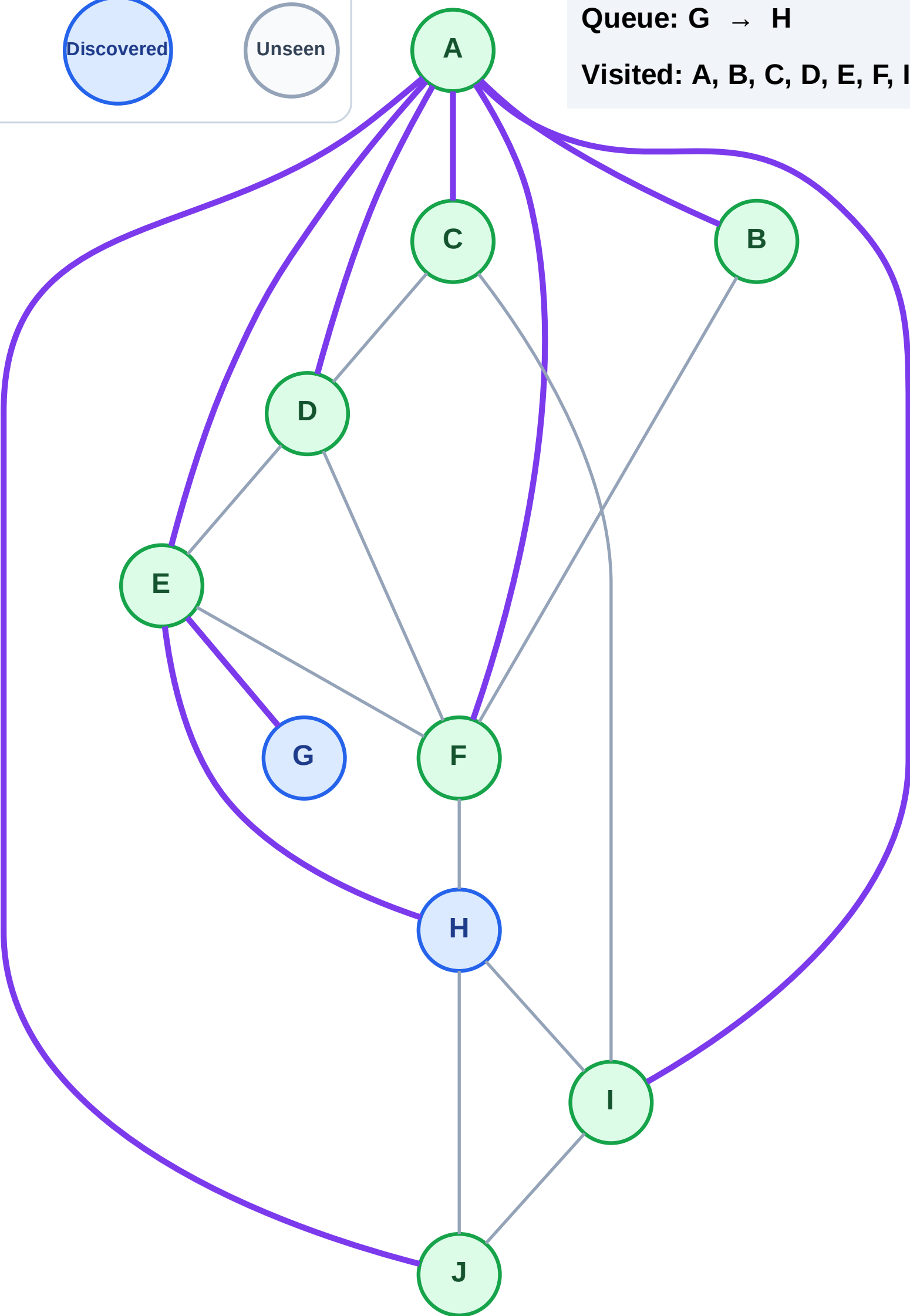
Step 17: No new neighbors from J

## Legend



Queue: G → H

Visited: A, B, C, D, E, F, I, J



# BFS Traversal

Step 18: Process node G

Legend

Complete

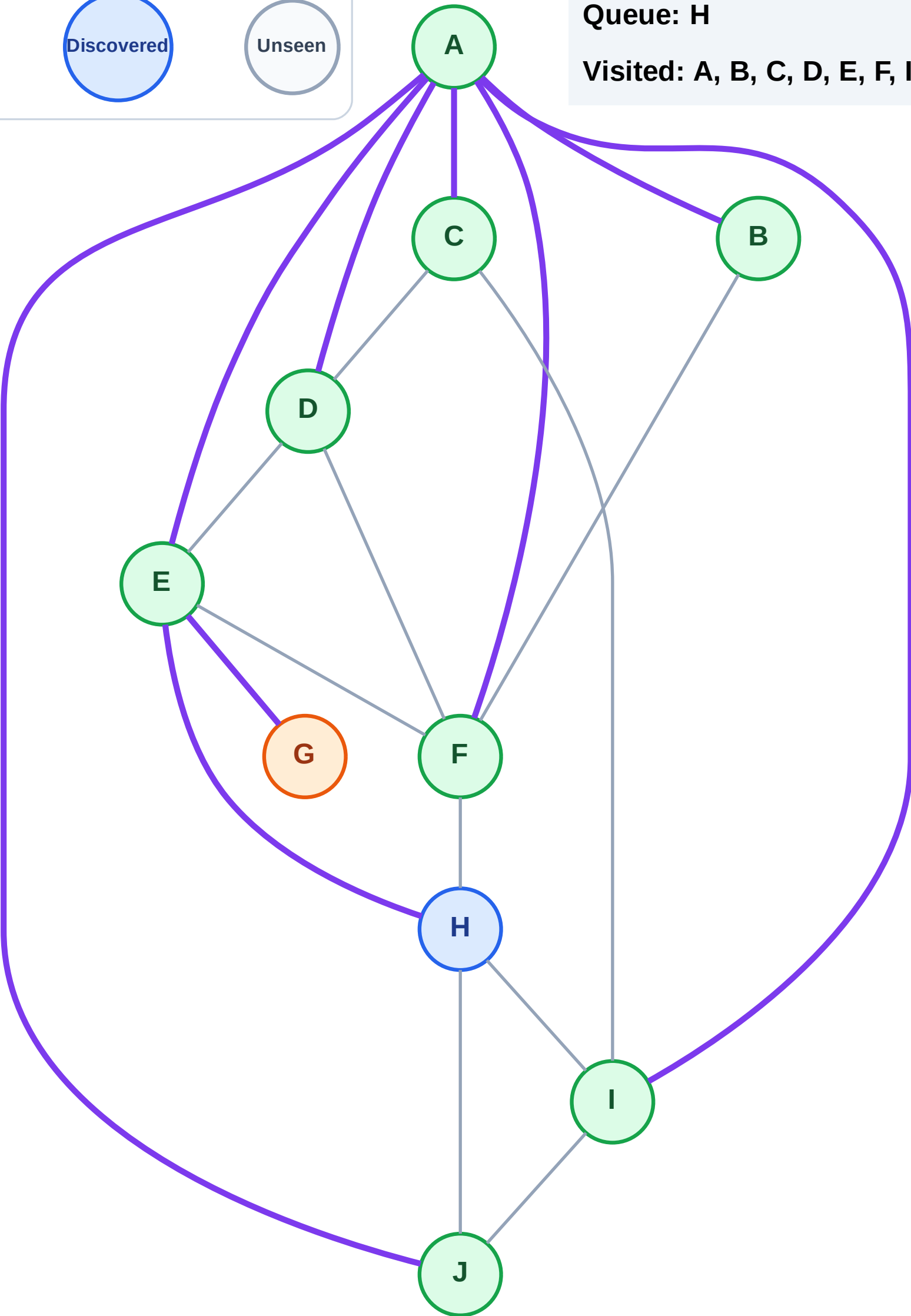
Processing

Discovered

Unseen

Queue: H

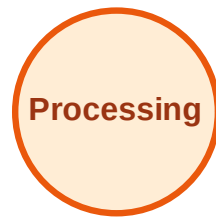
Visited: A, B, C, D, E, F, I, J



# BFS Traversal

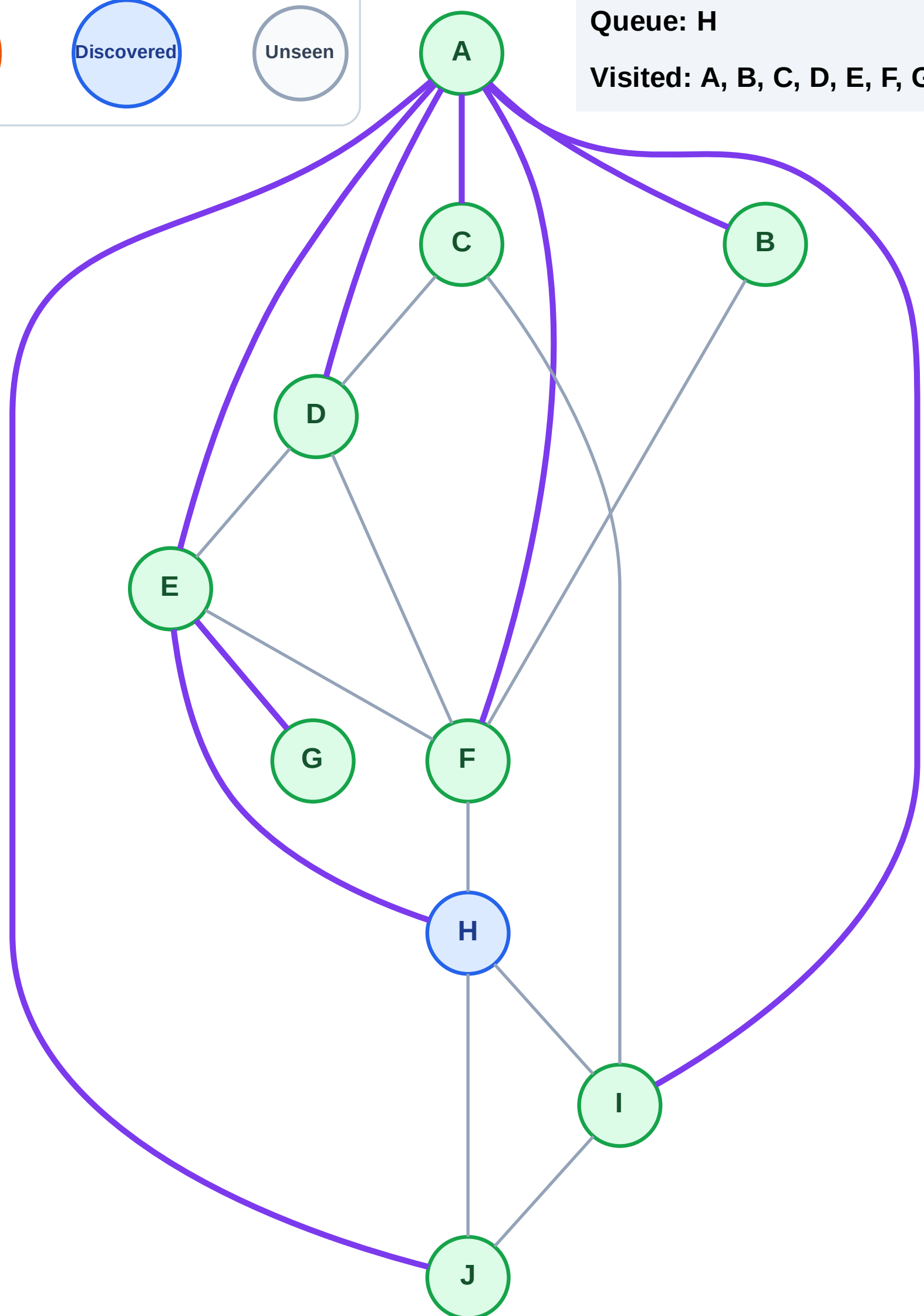
Step 19: No new neighbors from G

## Legend



Queue: H

Visited: A, B, C, D, E, F, G, I, J



# BFS Traversal

Step 20: Process node H

## Legend

Complete

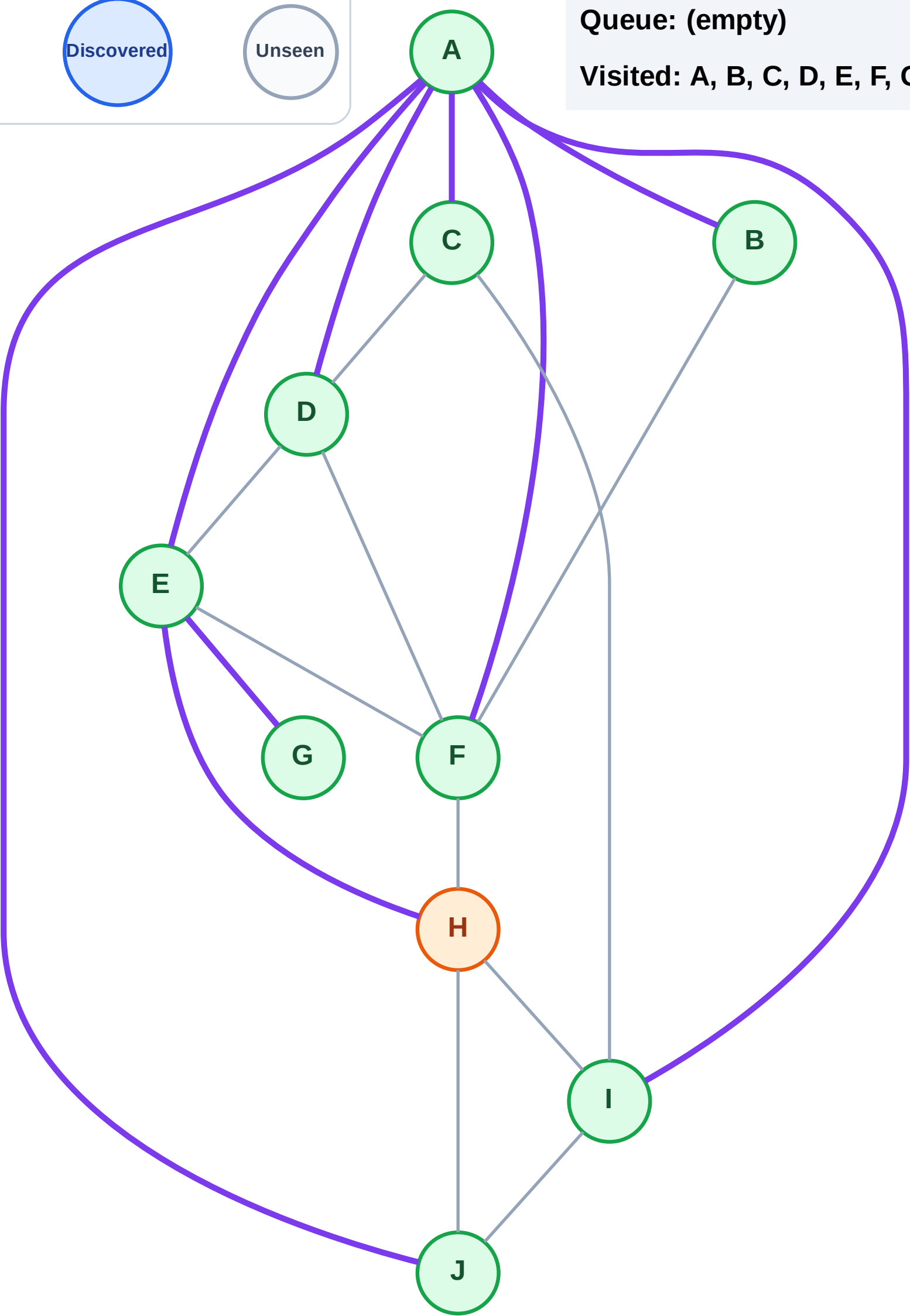
Processing

Discovered

Unseen

Queue: (empty)

Visited: A, B, C, D, E, F, G, I, J



# BFS Traversal

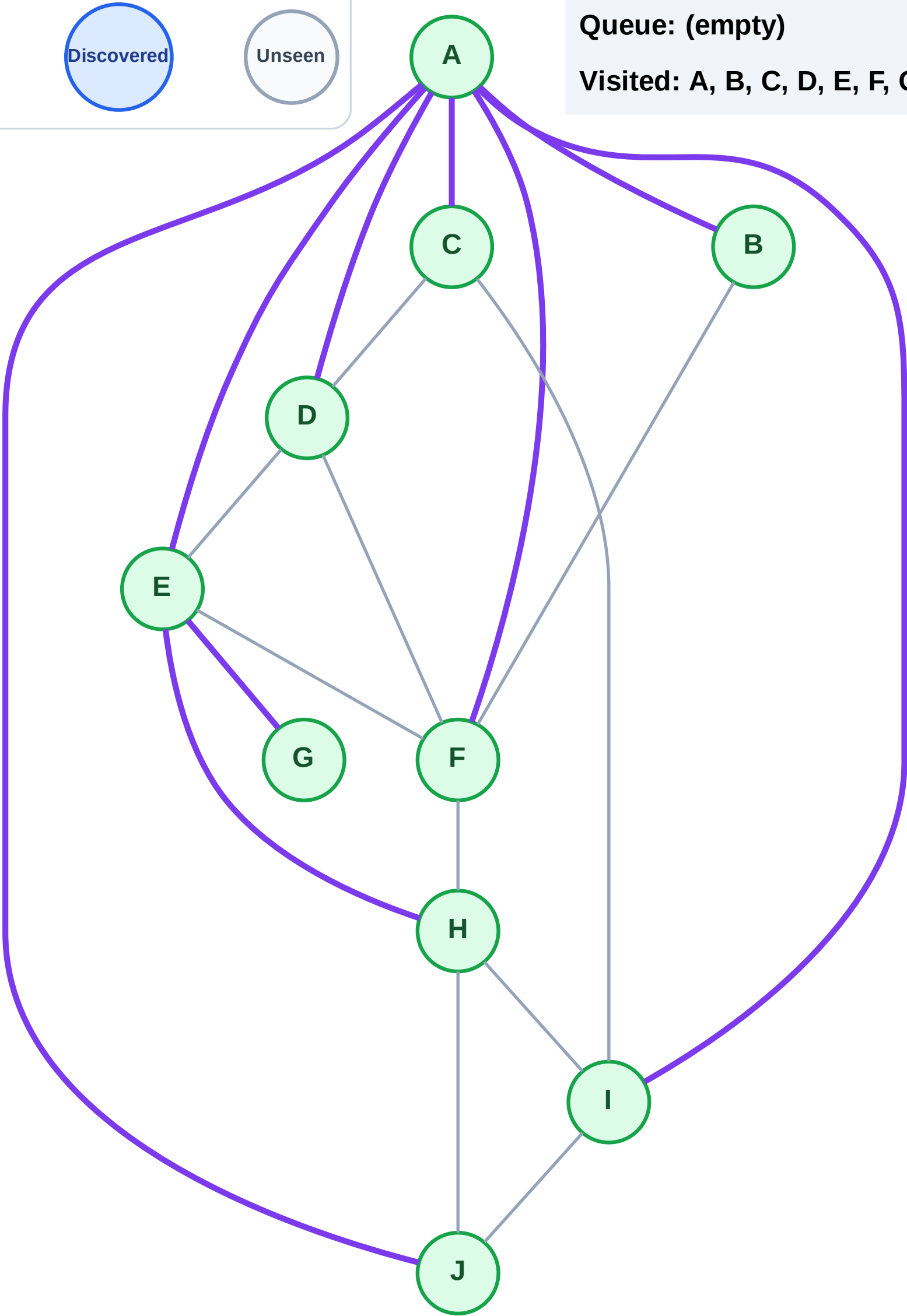
Step 21: No new neighbors from H

## Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



# DFS Traversal

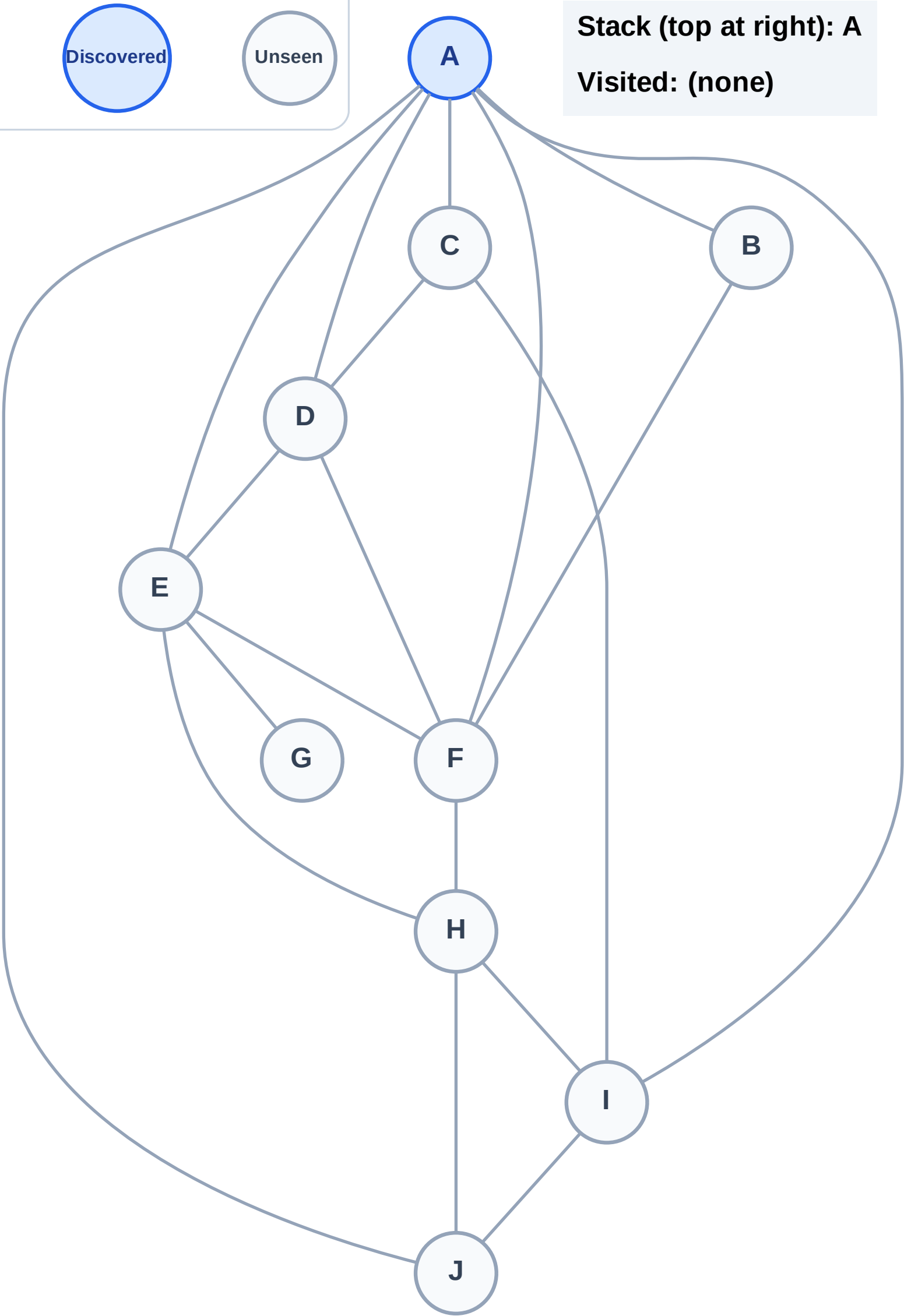
Step 1: Start at A

## Legend



Stack (top at right): A

Visited: (none)





# DFS Traversal

Step 2: Process node A

Legend

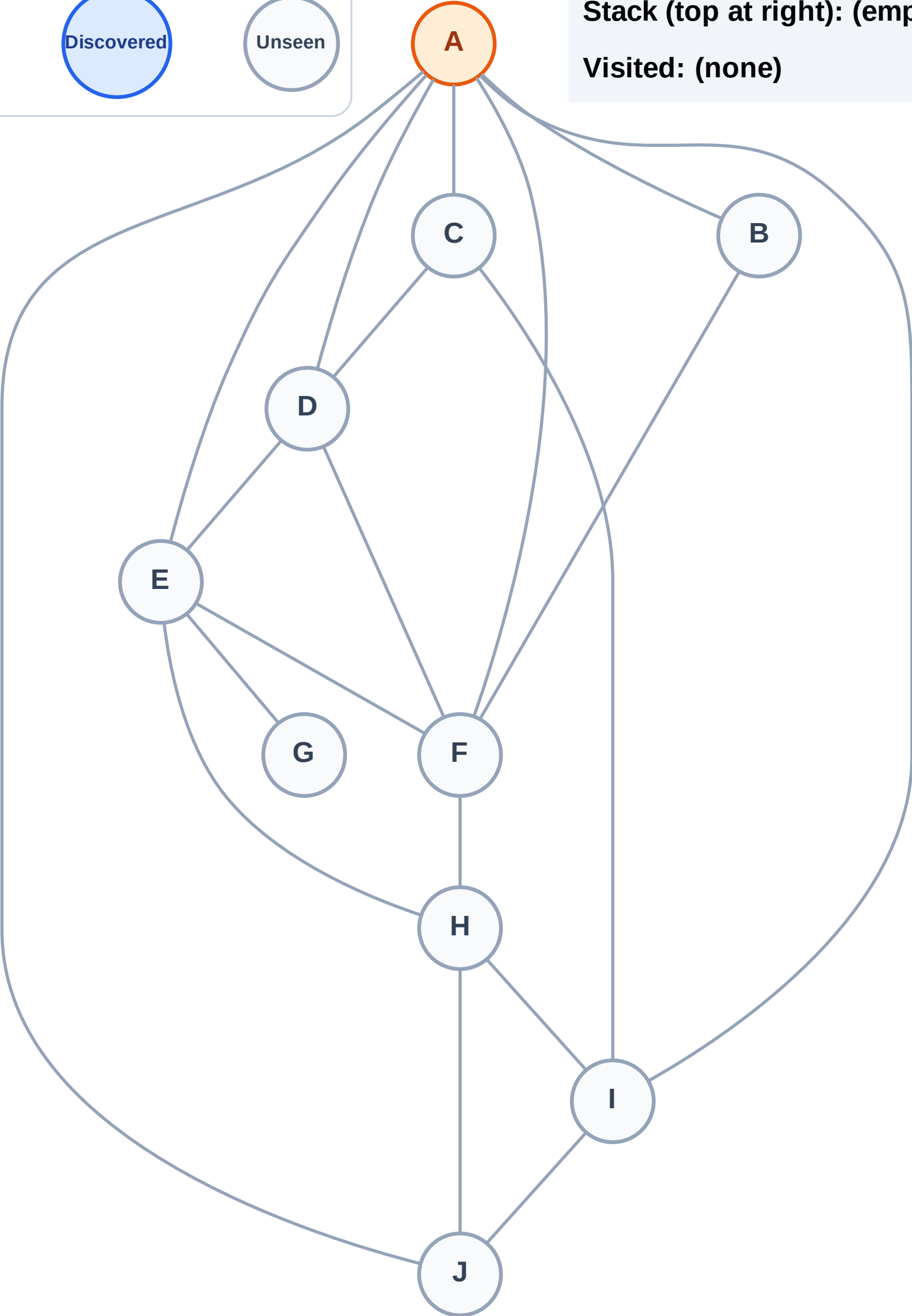
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)  
Visited: (none)



# DFS Traversal

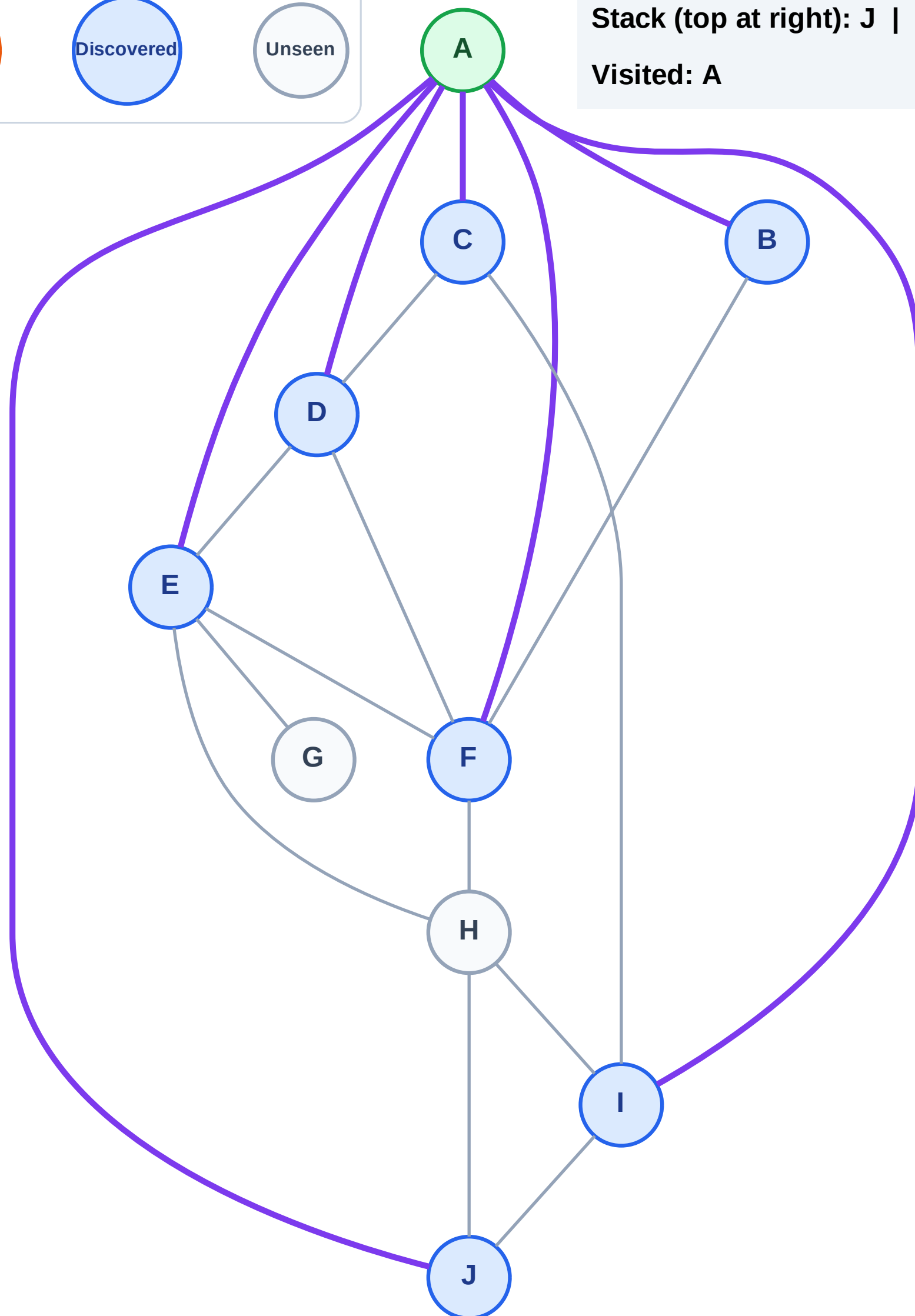
Step 3: Discover J, I, F, E, D, C, B from A

## Legend



Stack (top at right): J | I | F | E | D | C | B

Visited: A

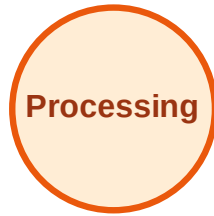




# DFS Traversal

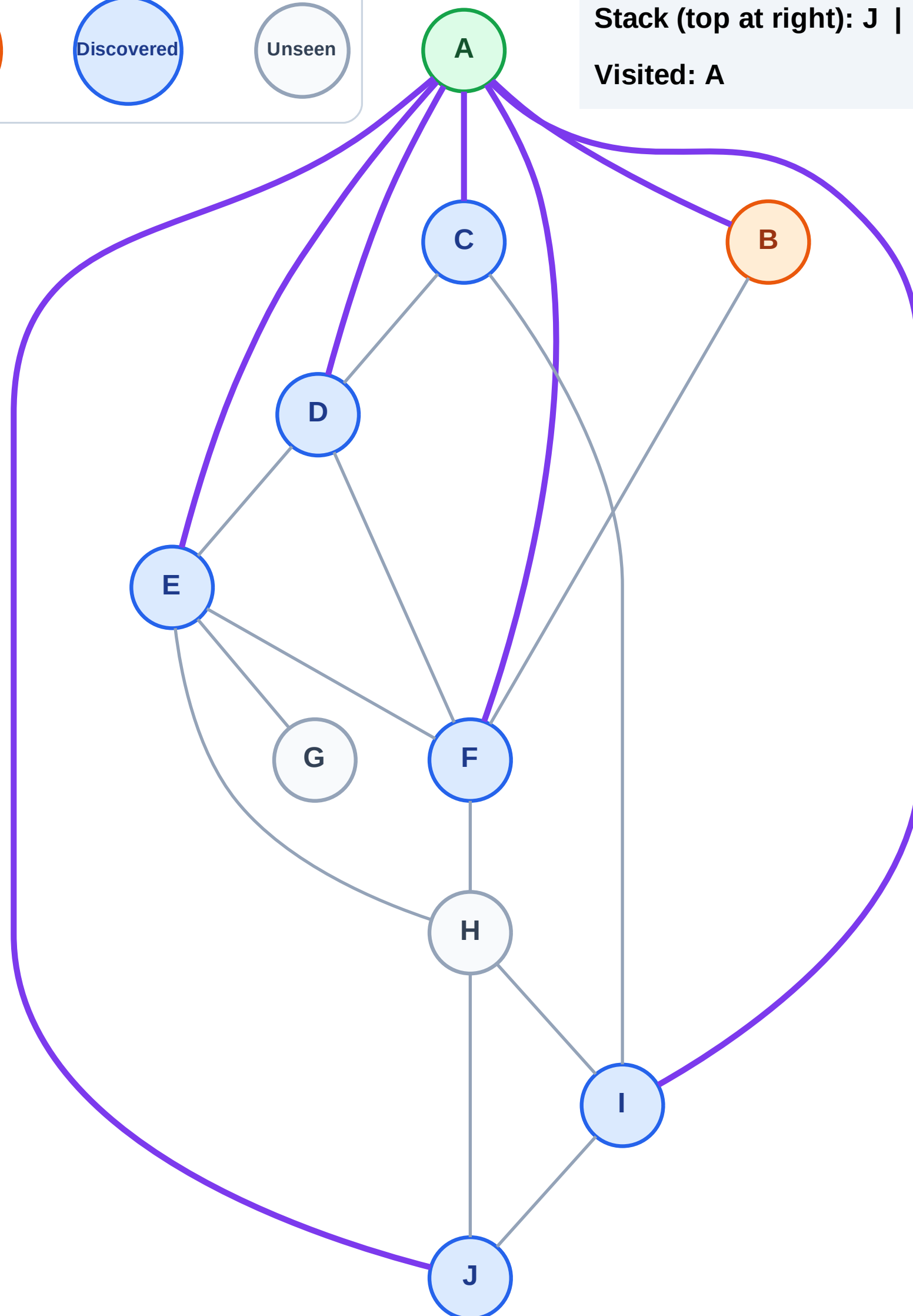
Step 4: Process node B

## Legend



Stack (top at right): J | I | F | E | D | C

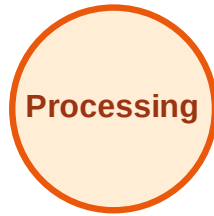
Visited: A



# DFS Traversal

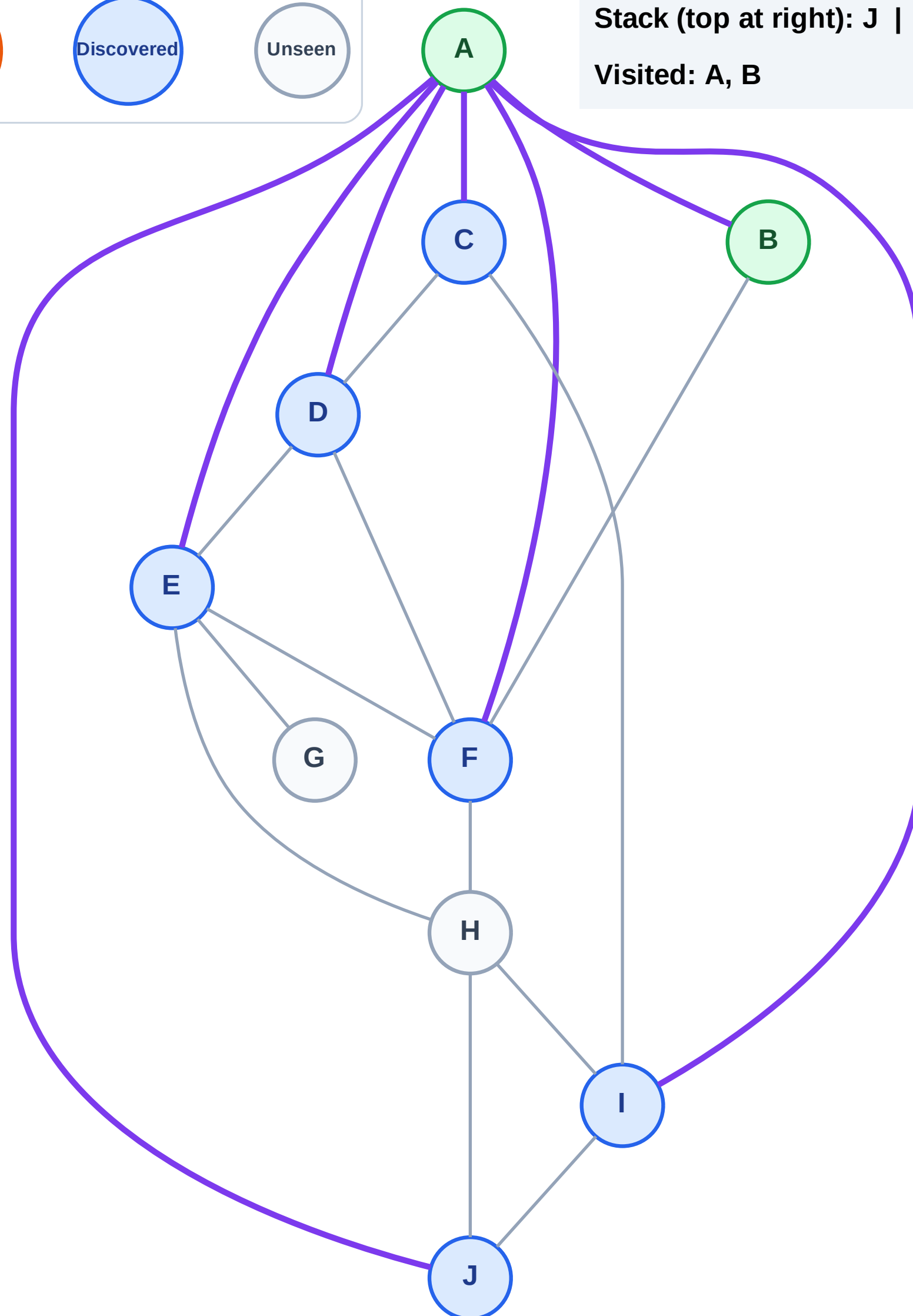
Step 5: No new neighbors from B

## Legend



Stack (top at right): J | I | F | E | D | C

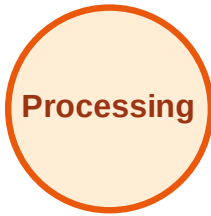
Visited: A, B



# DFS Traversal

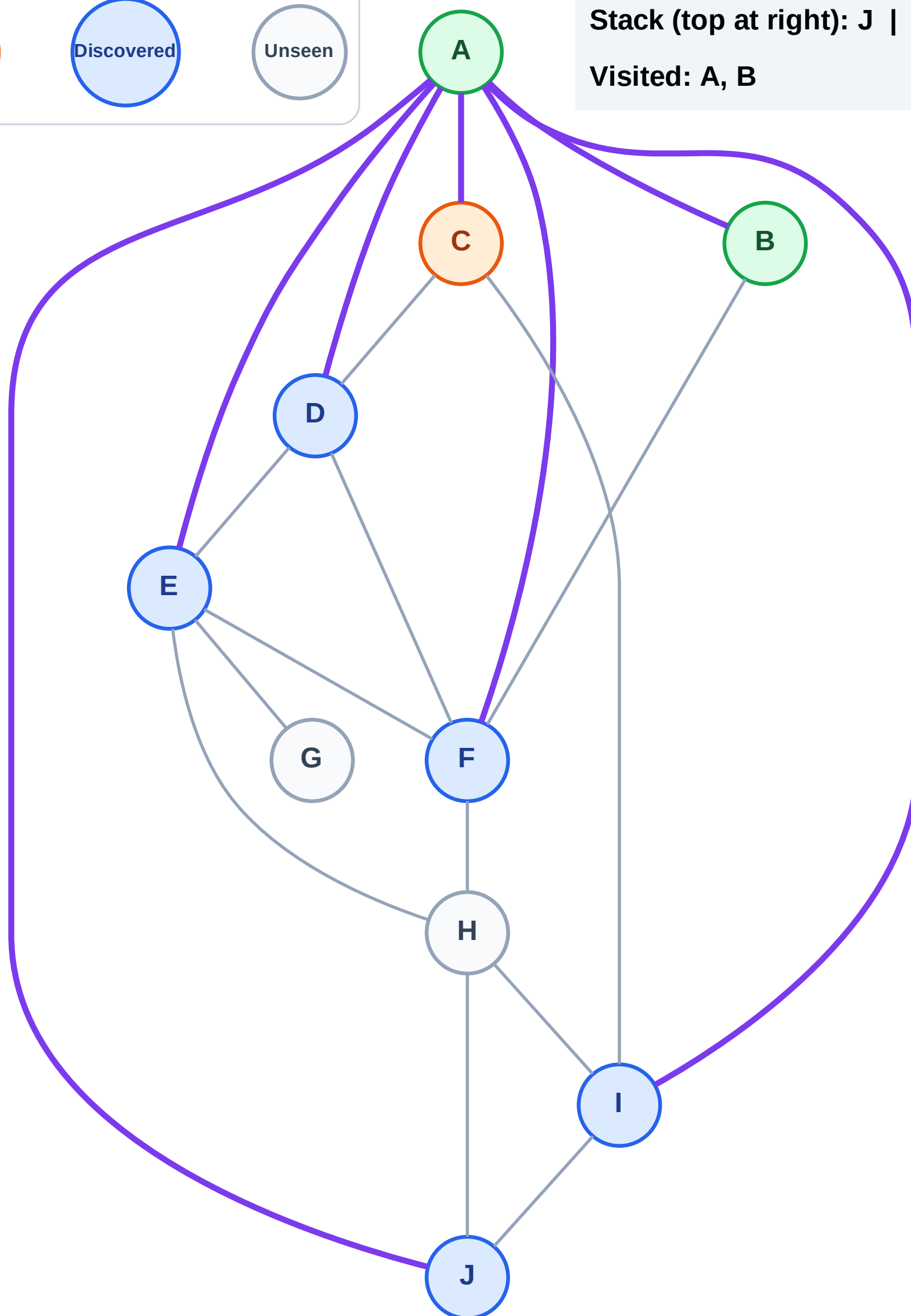
Step 6: Process node C

## Legend



Stack (top at right): J | I | F | E | D

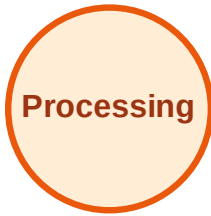
Visited: A, B



# DFS Traversal

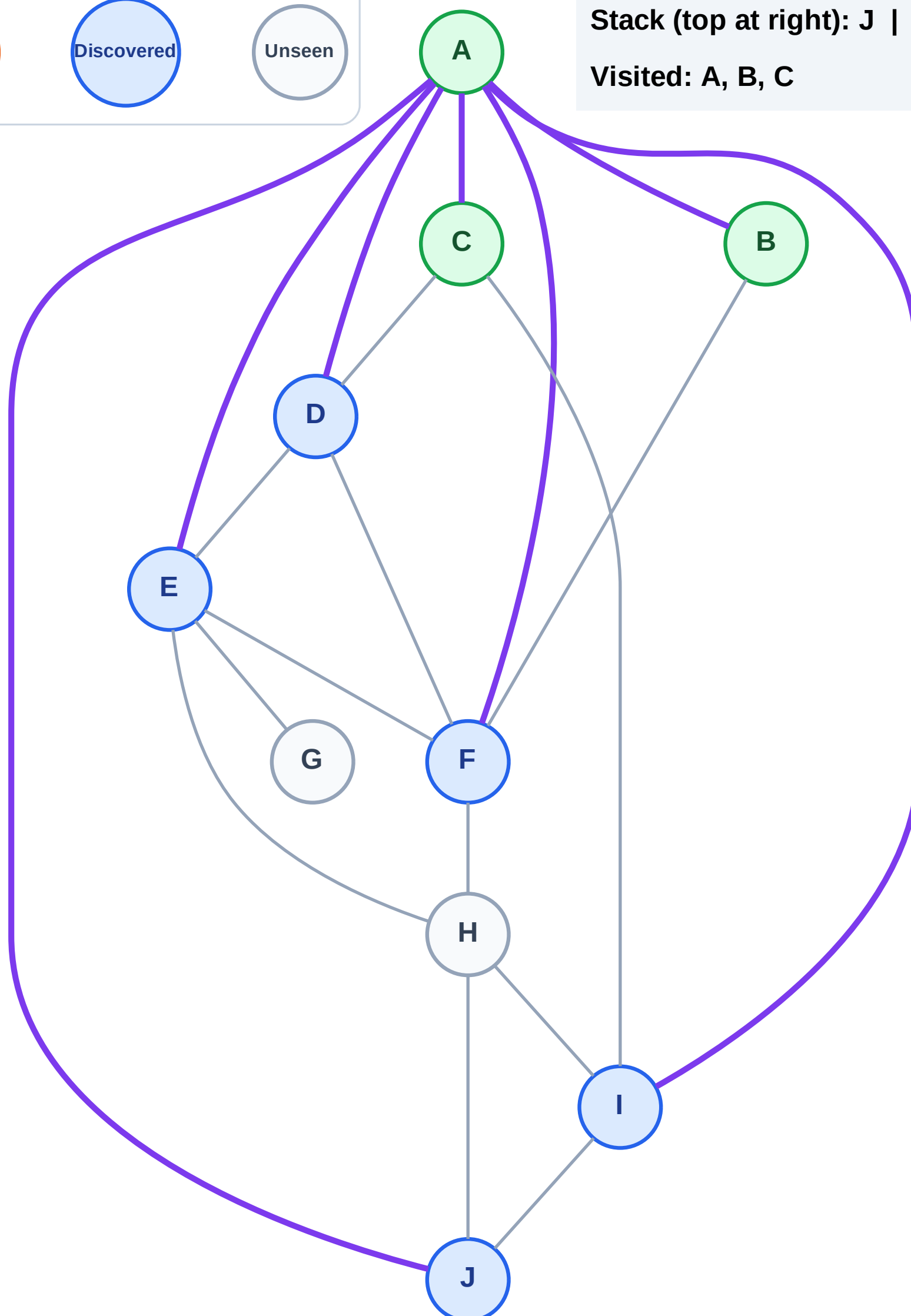
Step 7: No new neighbors from C

## Legend



Stack (top at right): J | I | F | E | D

Visited: A, B, C



# DFS Traversal

Step 8: Process node D

Legend

Complete

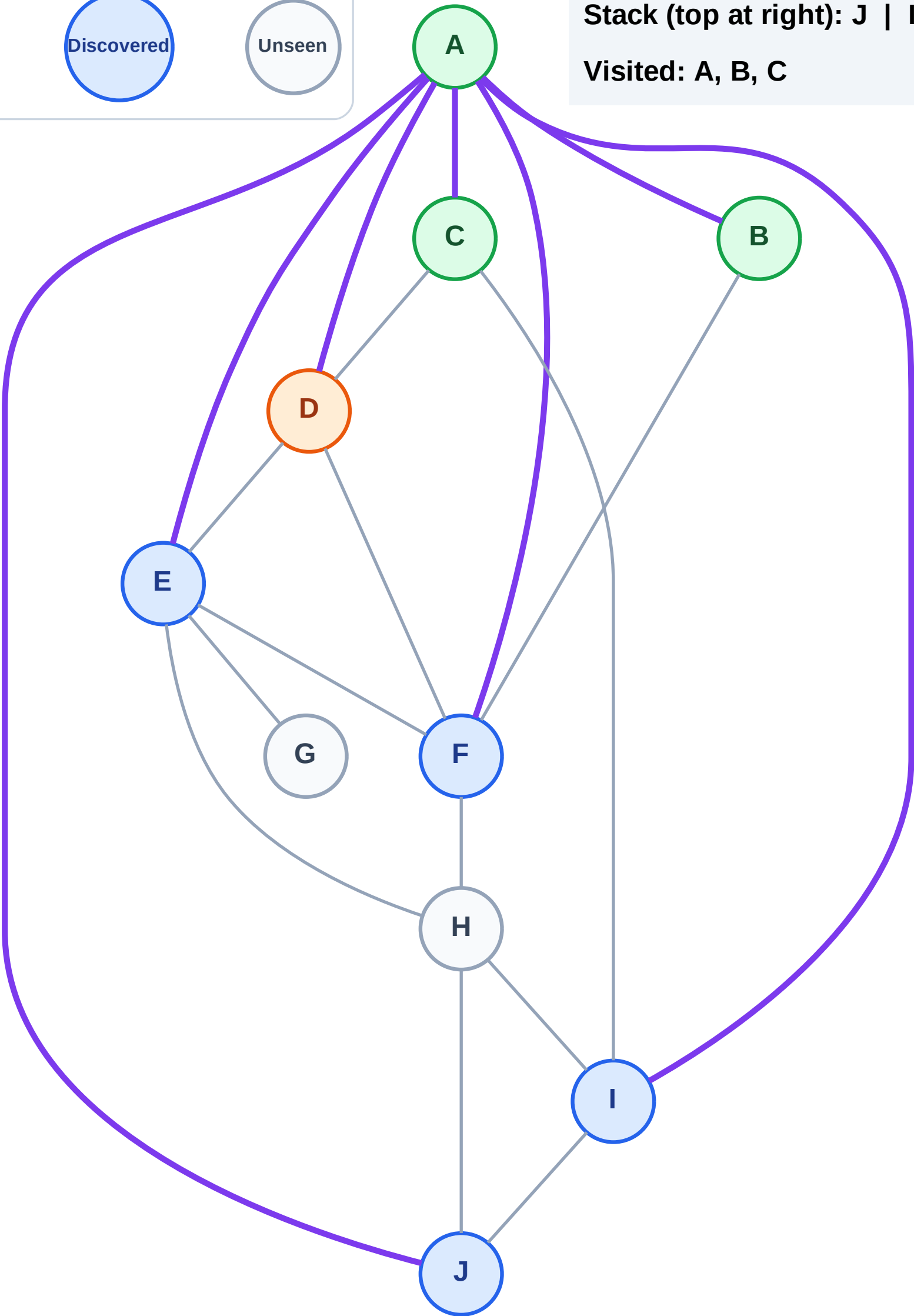
Processing

Discovered

Unseen

Stack (top at right): J | I | F | E

Visited: A, B, C





# DFS Traversal

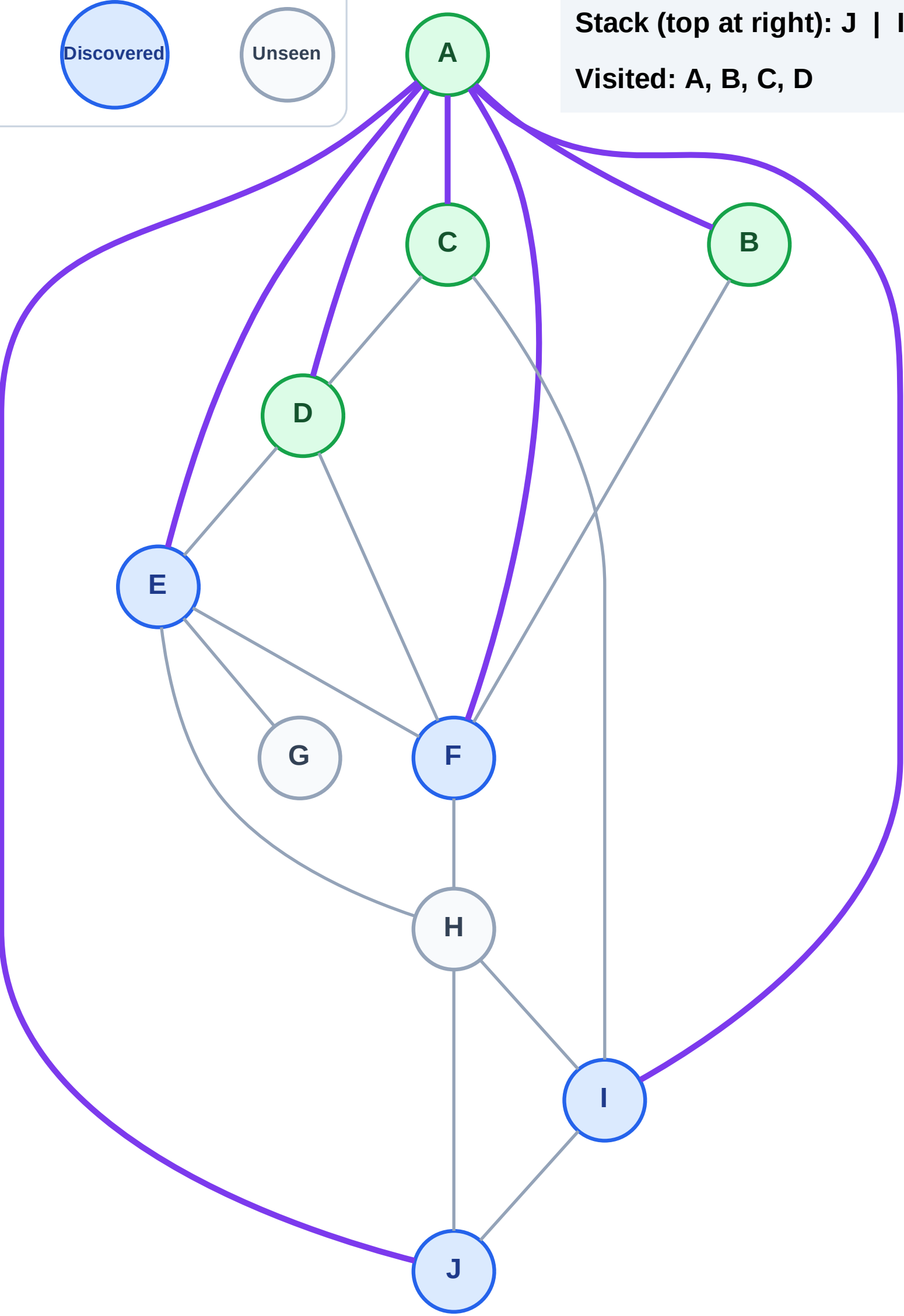
Step 9: No new neighbors from D

## Legend



Stack (top at right): J | I | F | E

Visited: A, B, C, D



# DFS Traversal

Step 10: Process node E

Legend

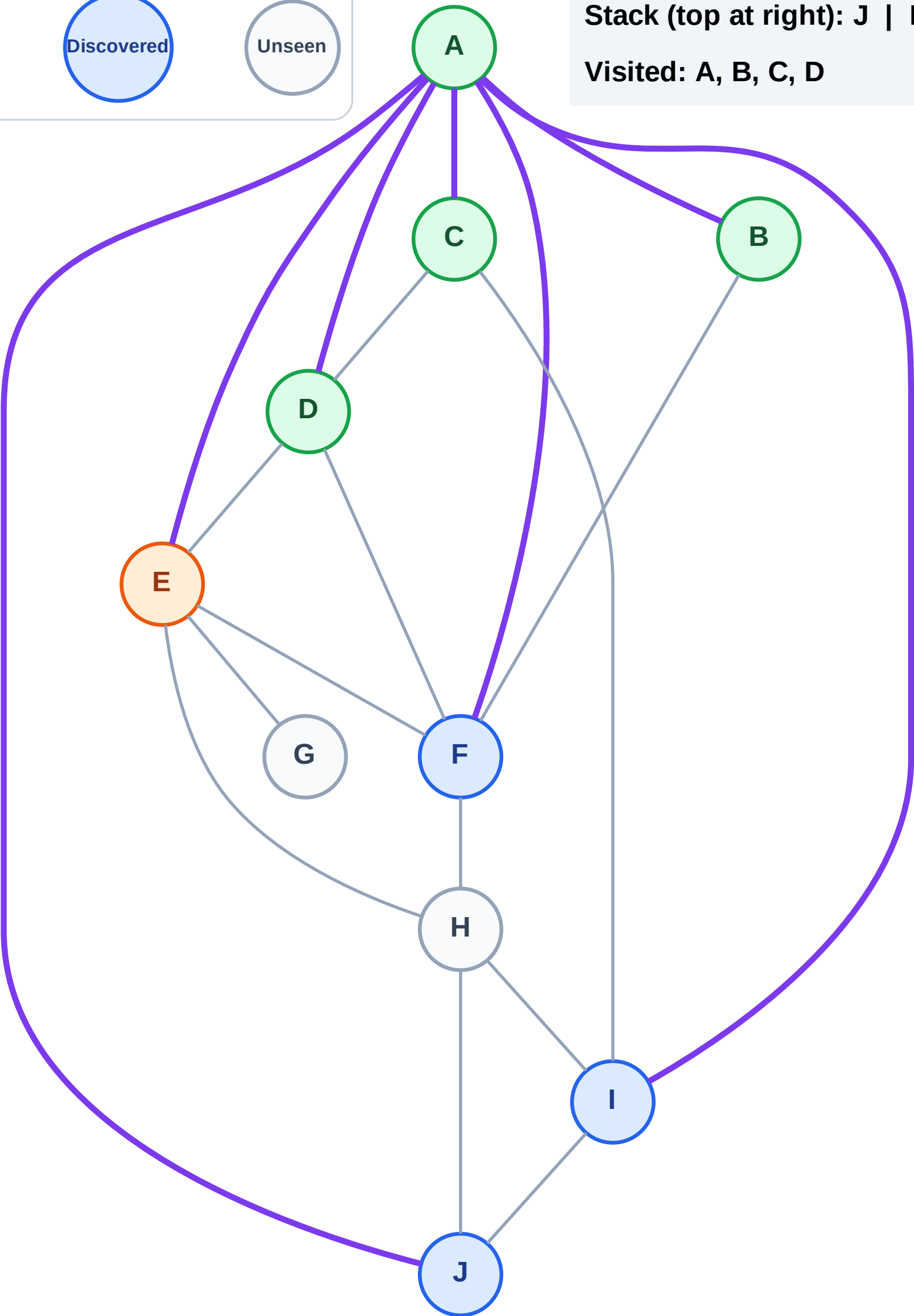
Complete

Processing

Discovered

Unseen

Stack (top at right): J | I | F  
Visited: A, B, C, D



# DFS Traversal

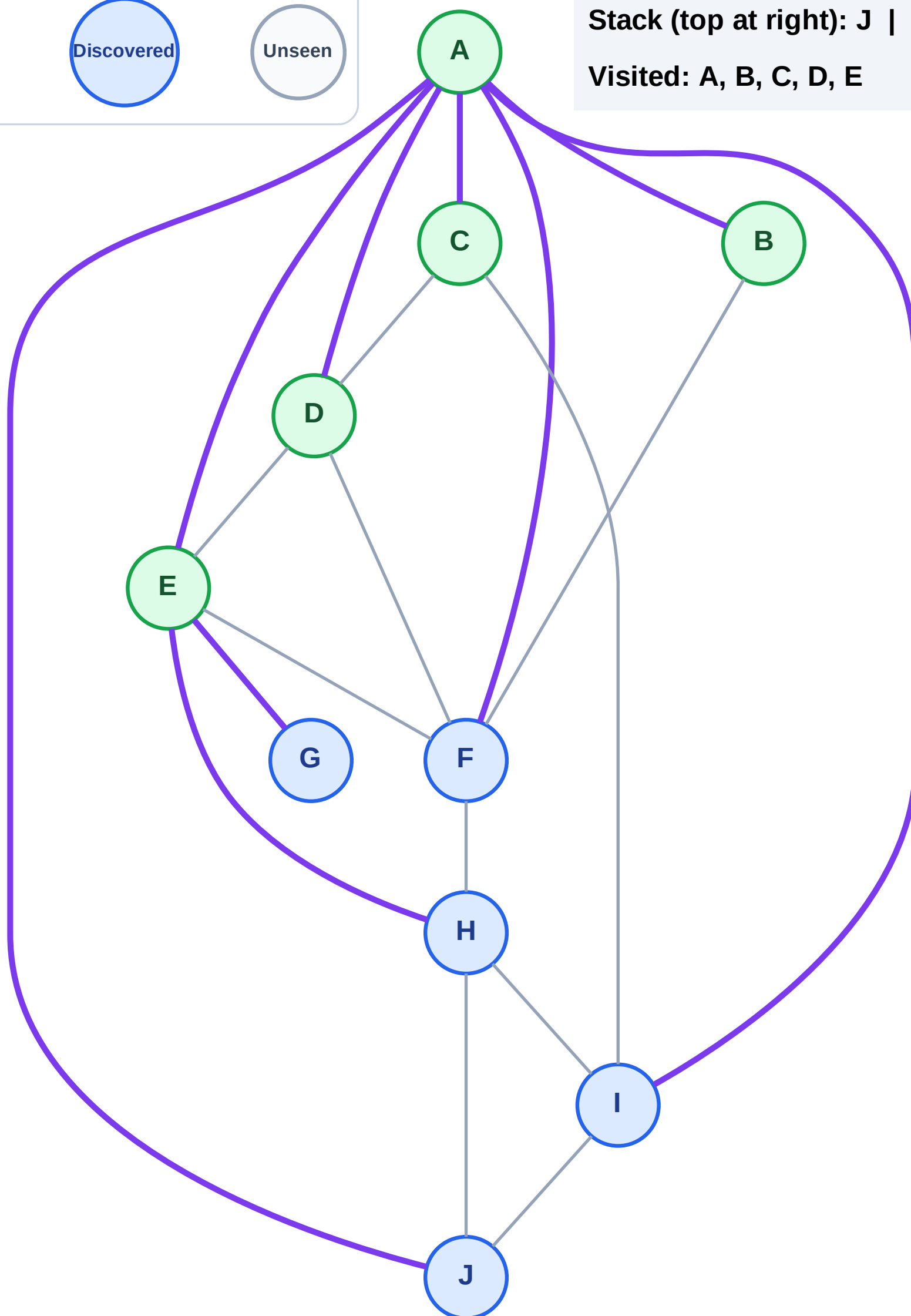
Step 11: Discover H, G from E

## Legend



Stack (top at right): J | I | F | H | G

Visited: A, B, C, D, E

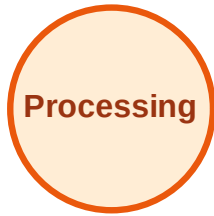




# DFS Traversal

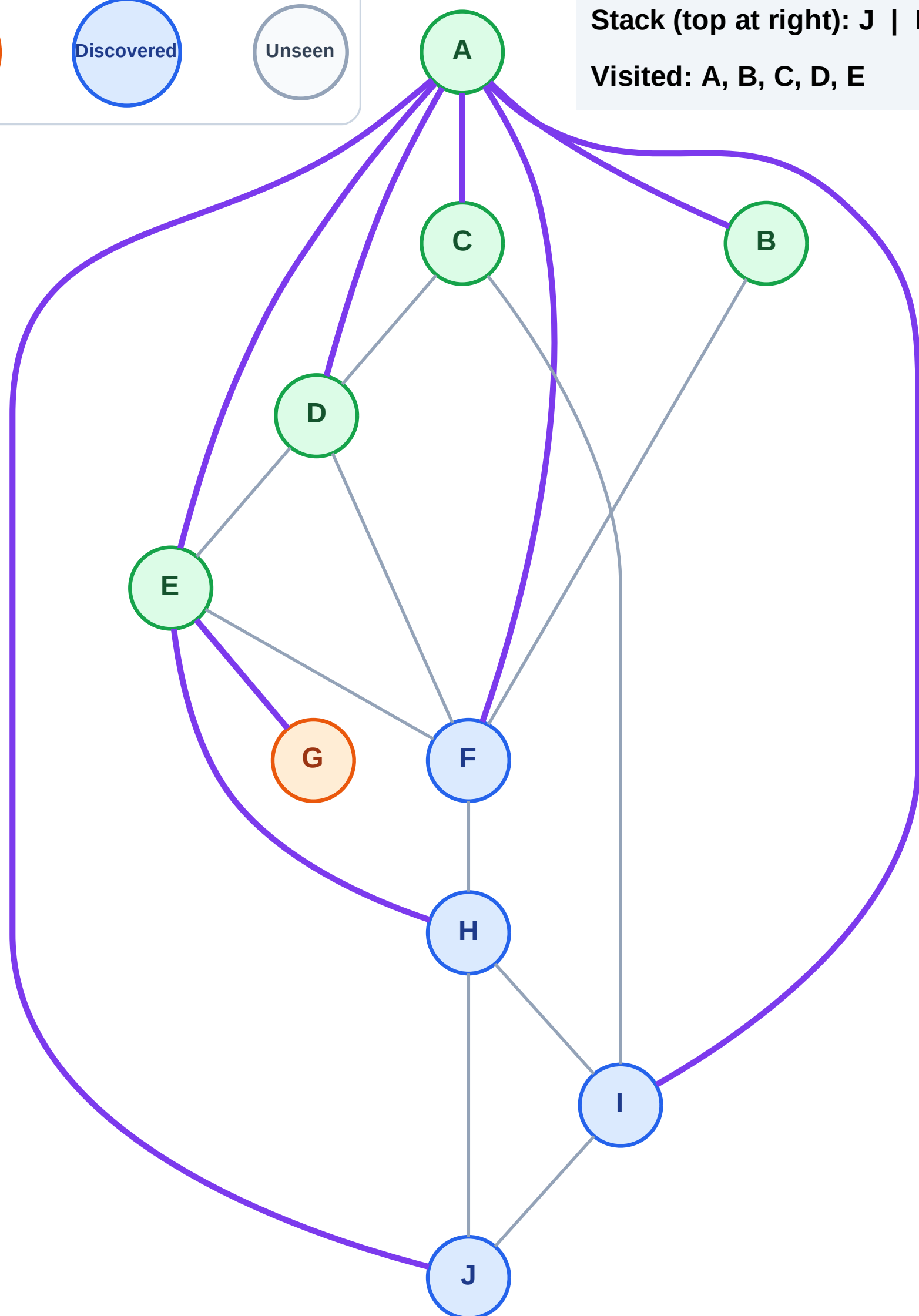
Step 12: Process node G

## Legend



Stack (top at right): J | I | F | H

Visited: A, B, C, D, E



# DFS Traversal

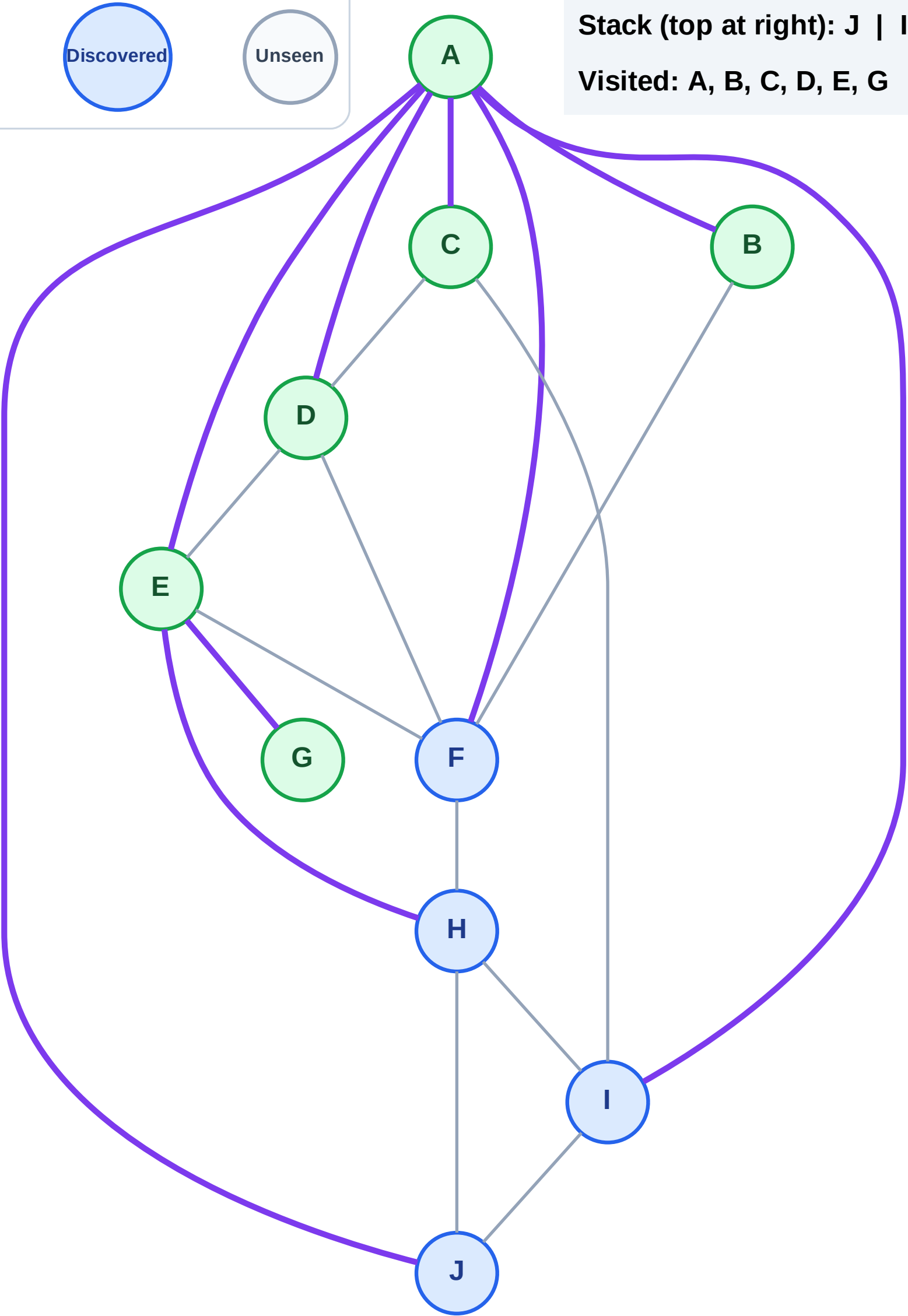
Step 13: No new neighbors from G

## Legend



Stack (top at right): J | I | F | H

Visited: A, B, C, D, E, G



# DFS Traversal

Step 14: Process node H

## Legend

Complete

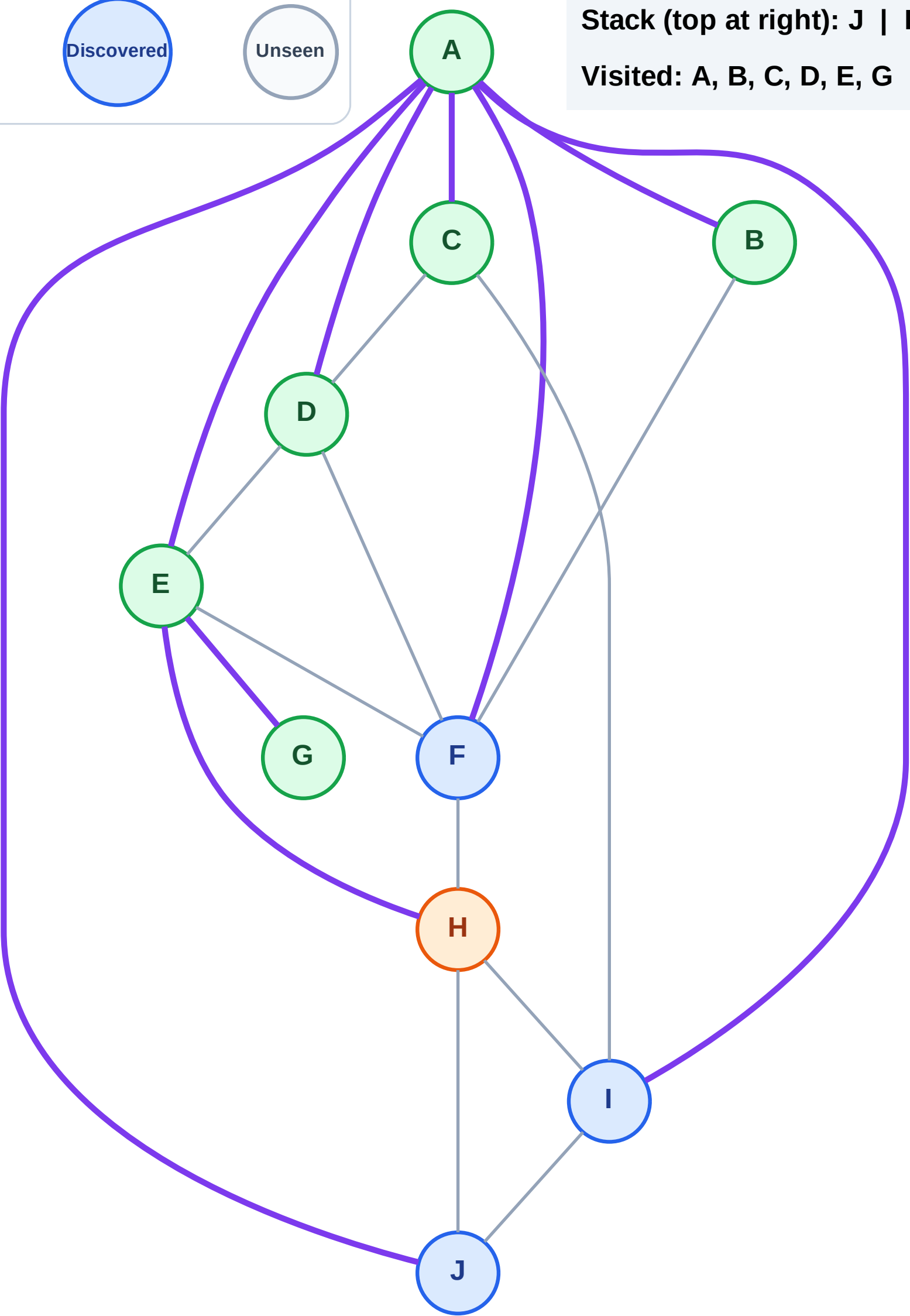
Processing

Discovered

Unseen

Stack (top at right): J | I | F

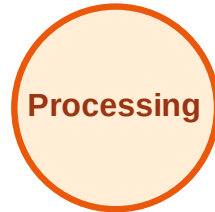
Visited: A, B, C, D, E, G



# DFS Traversal

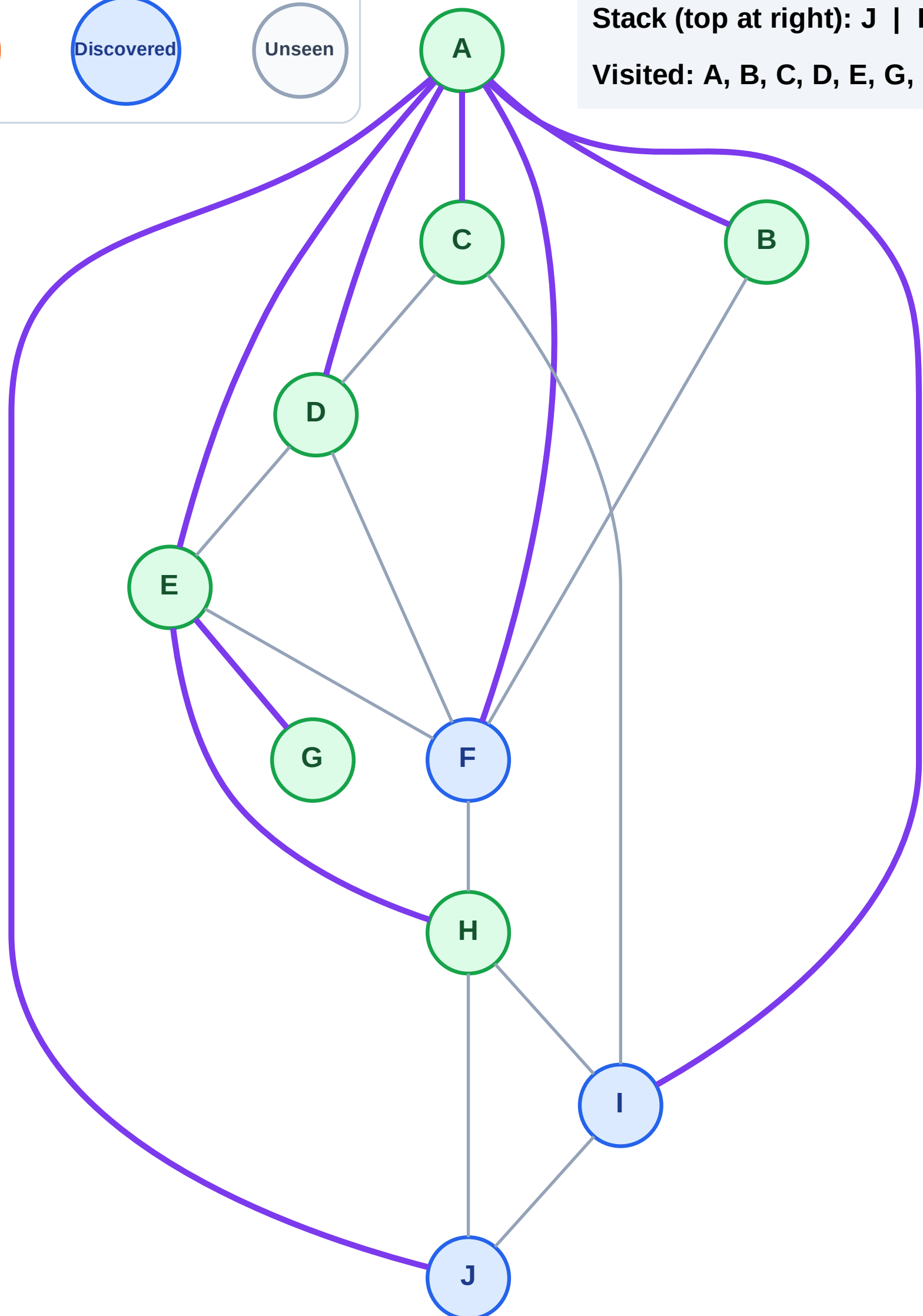
Step 15: No new neighbors from H

## Legend



Stack (top at right): J | I | F

Visited: A, B, C, D, E, G, H



# DFS Traversal

Step 16: Process node F

Legend

Complete

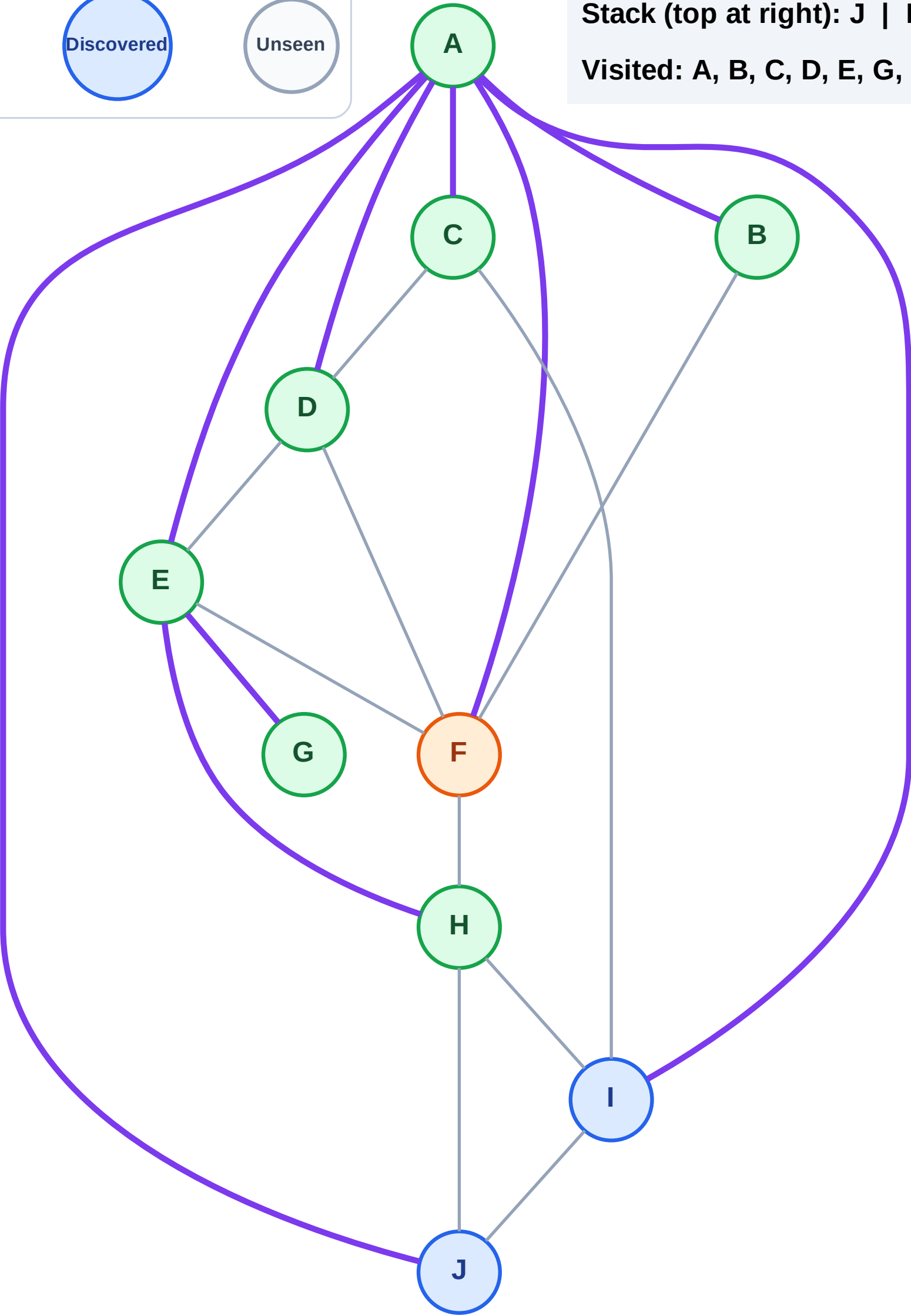
Processing

Discovered

Unseen

Stack (top at right): J | I

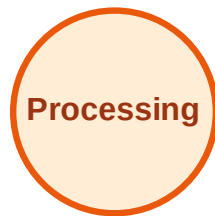
Visited: A, B, C, D, E, G, H



# DFS Traversal

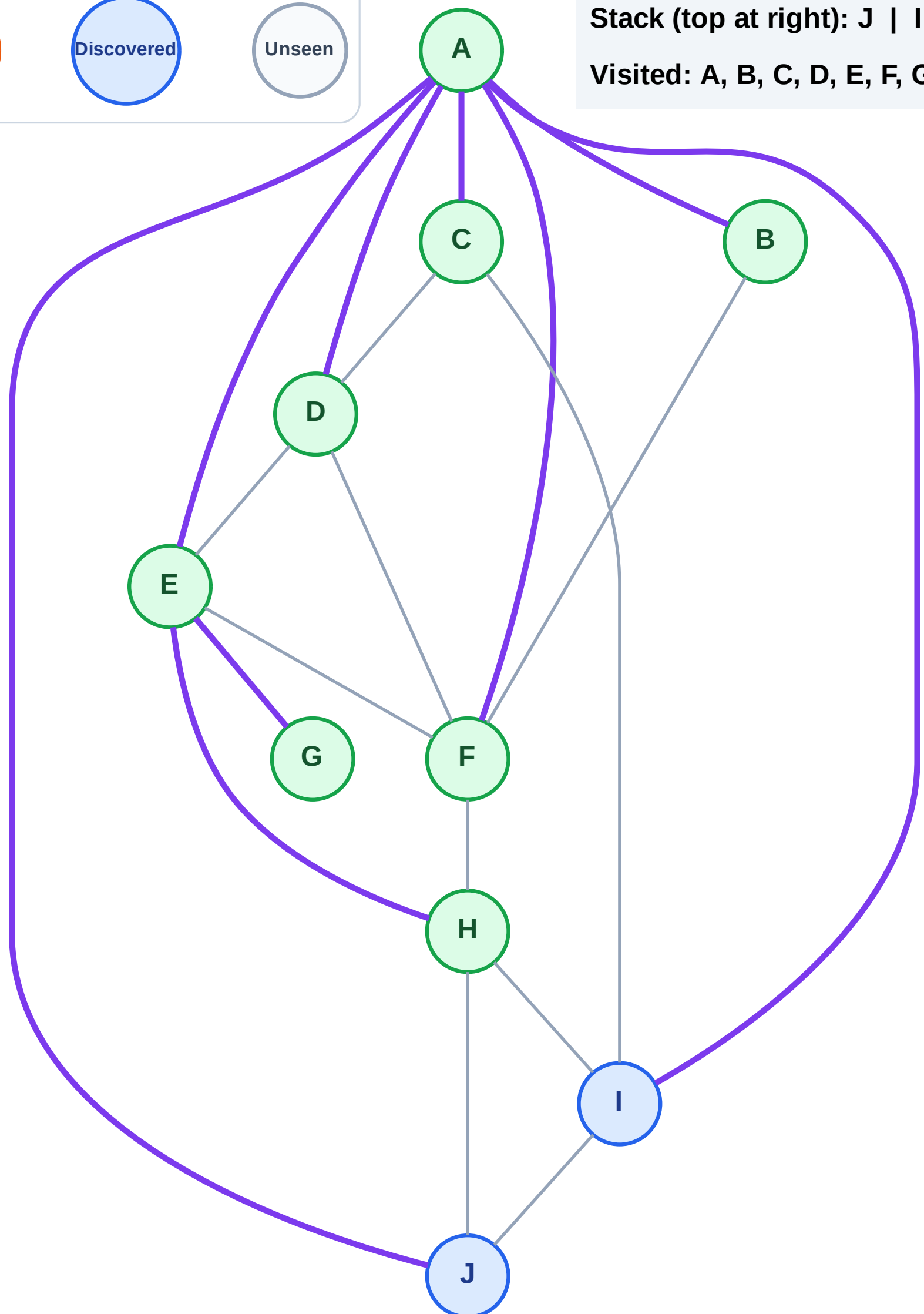
Step 17: No new neighbors from F

## Legend



Stack (top at right): J | I

Visited: A, B, C, D, E, F, G, H

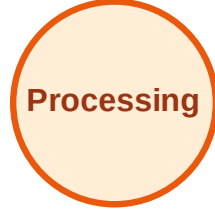




# DFS Traversal

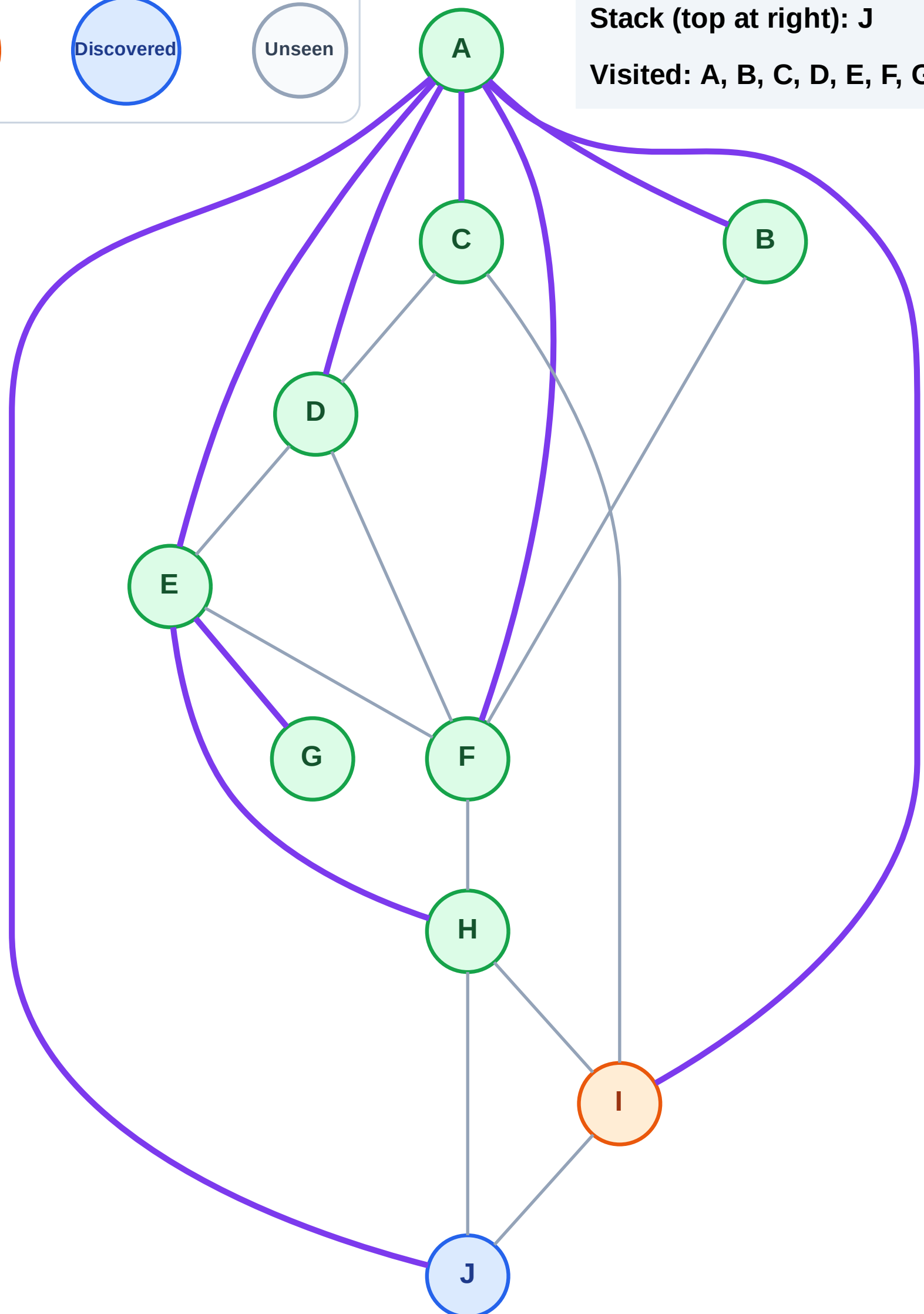
Step 18: Process node I

## Legend



Stack (top at right): J

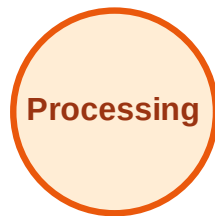
Visited: A, B, C, D, E, F, G, H



# DFS Traversal

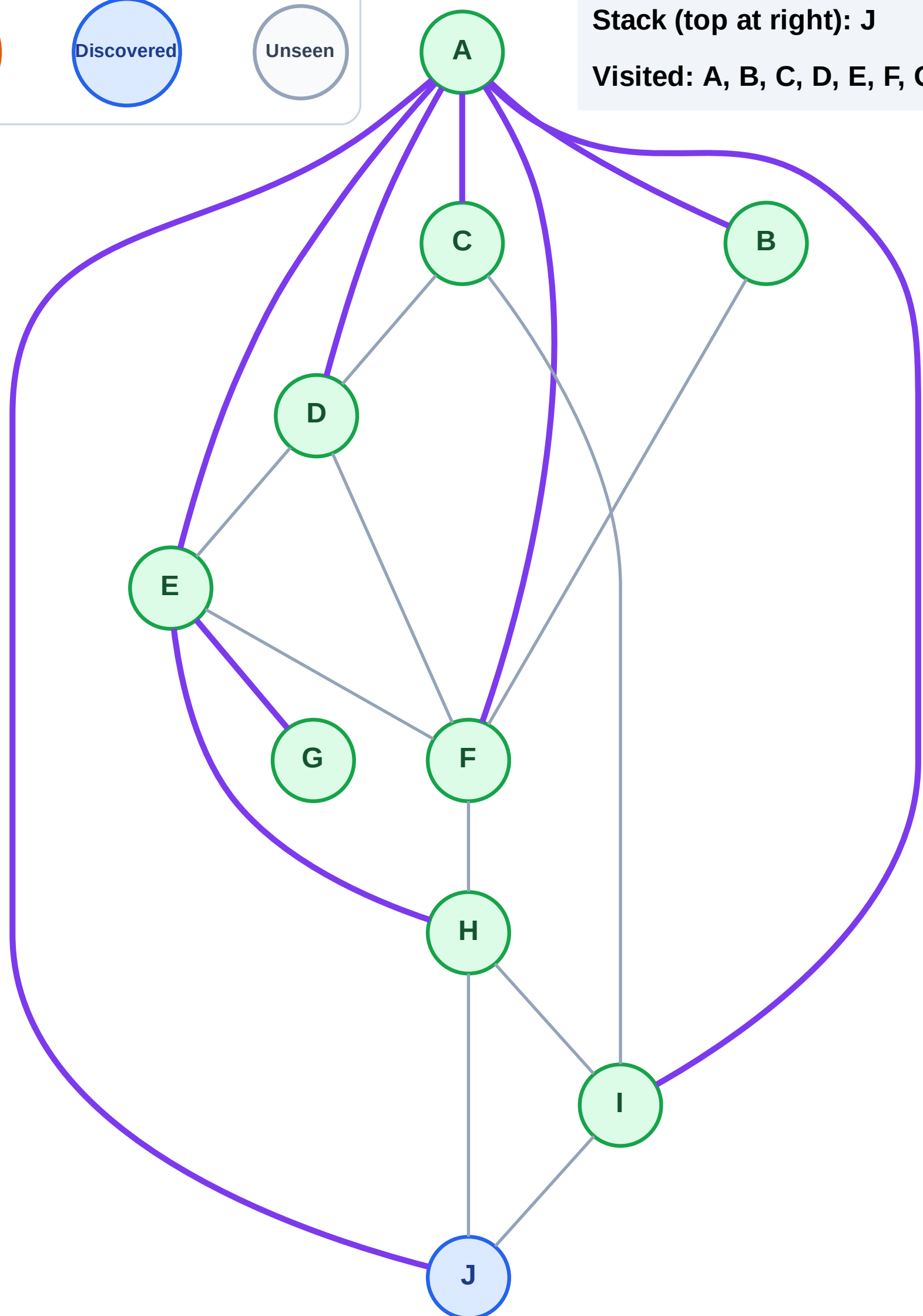
Step 19: No new neighbors from I

## Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I

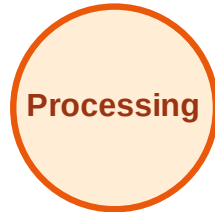




# DFS Traversal

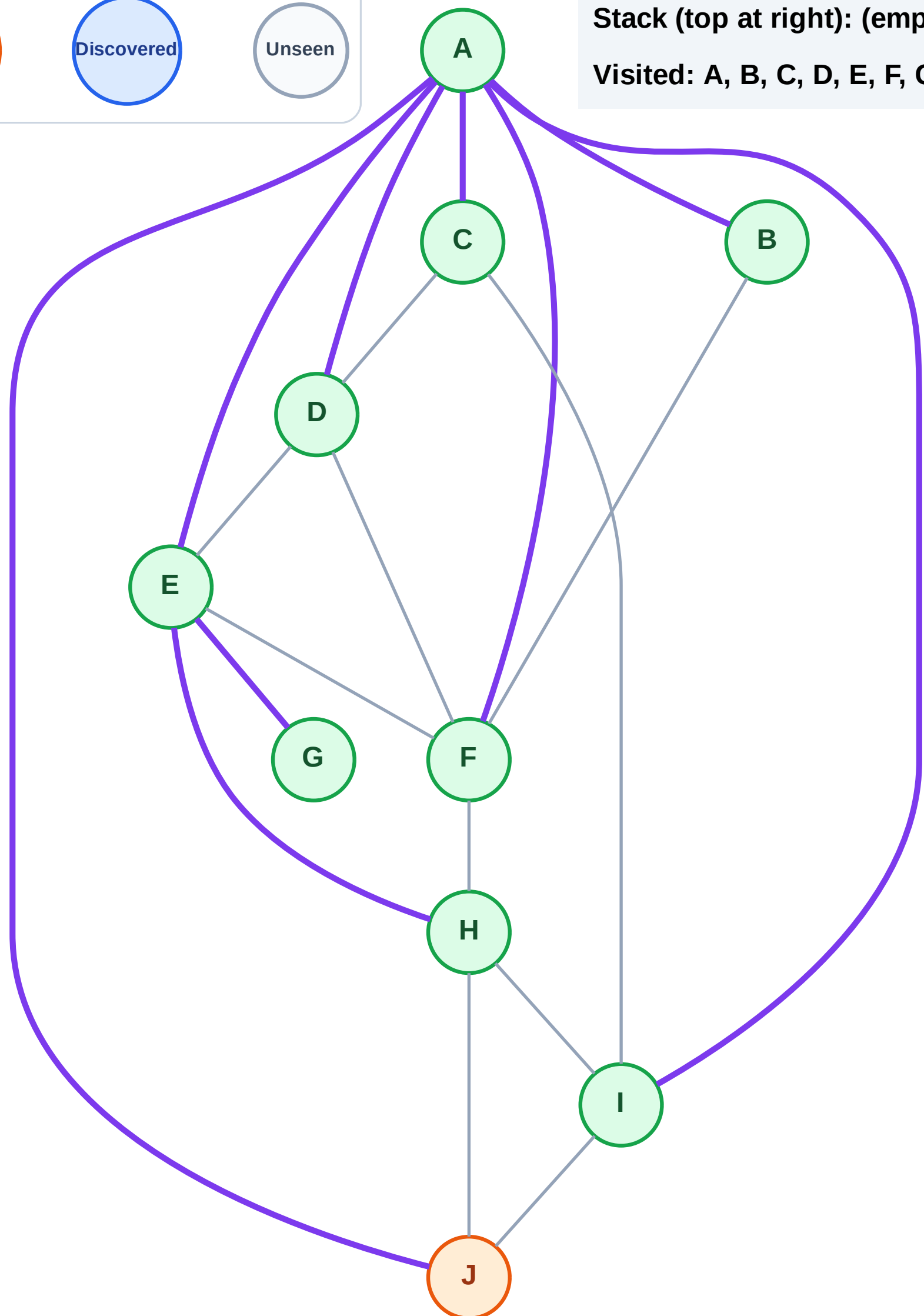
Step 20: Process node J

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I



# DFS Traversal

Step 21: No new neighbors from J

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

